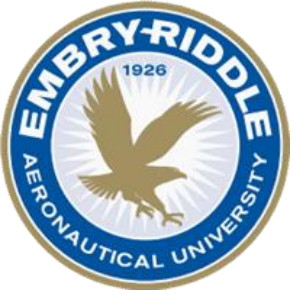


Introduction to Computer Networks



Introduction

Goal:

Get “feel,” “big picture,”
introduction to terminology

Overview/roadmap:

What *is* the Internet? What *is* a protocol?

Network edge: hosts, access network,
physical media

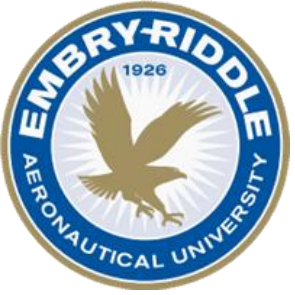
Network core: packet/circuit switching,
internet structure

Performance: loss, delay, throughput

Protocol layers, service models

Security

History



Roadmap

What ***is*** the Internet?

What *is* a protocol?

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physical media

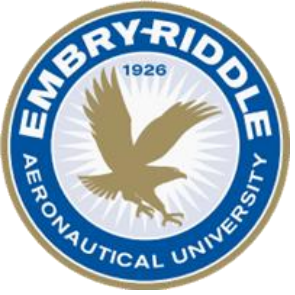
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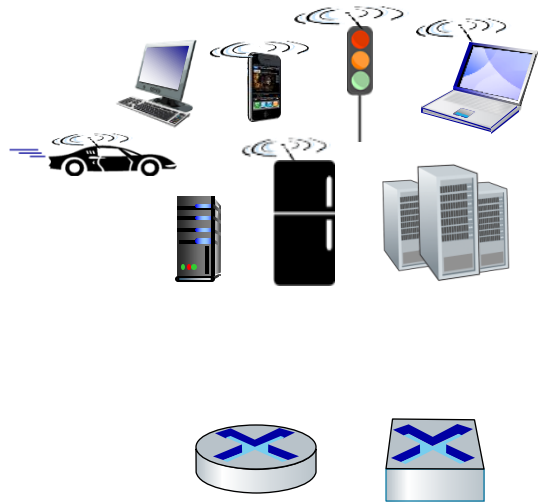
Security

Protocol layers, service models

History



The Internet

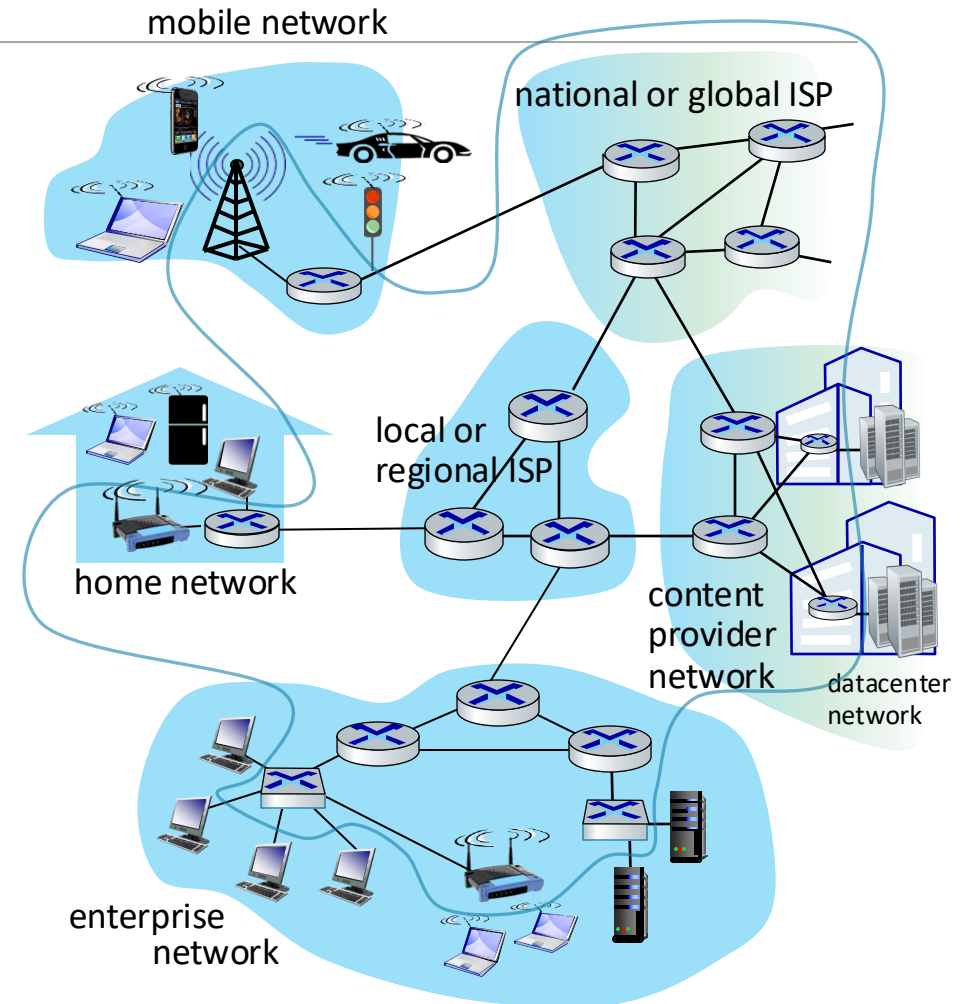


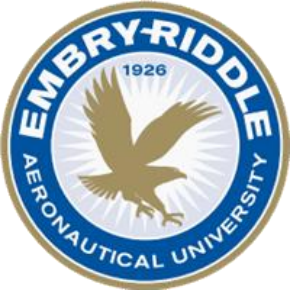
Billions of connected computing *devices*:

- *hosts* = end systems
- running network *apps/programs* at the Internet's "edge"

Packet switches: forward packets (chunks of data)

- *routers, switches*





The Internet

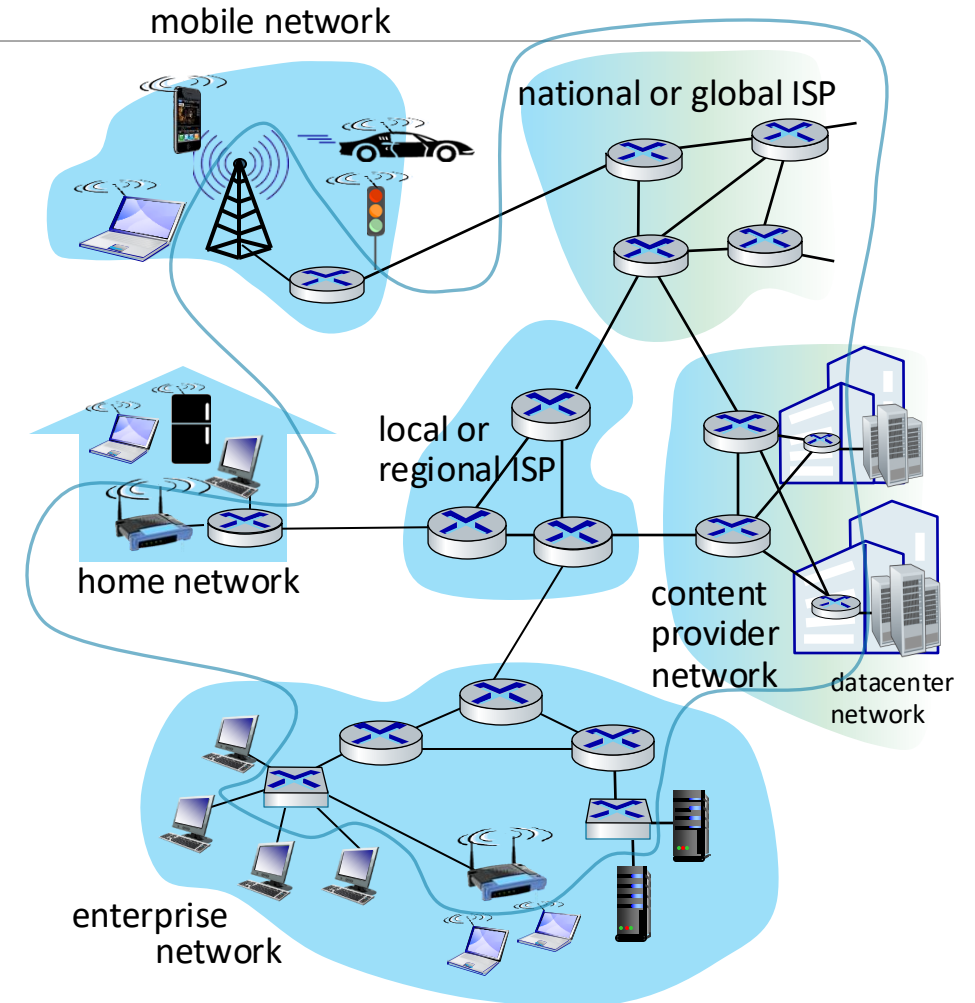


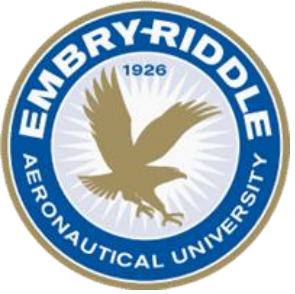
Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization





“Fun” Internet-connected devices



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt:
monitor energy use



Security Camera



Slingbox: remote control cable TV



Web-enabled toaster +
weather forecaster



AR devices



cars



bikes



scooters



Internet phones



Gaming devices

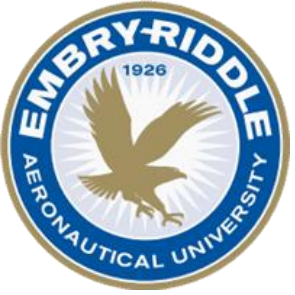


sensorized,
bed
mattress



Fitbit

Others?



The Internet: a “nuts and bolts” view

Internet: “network of networks”

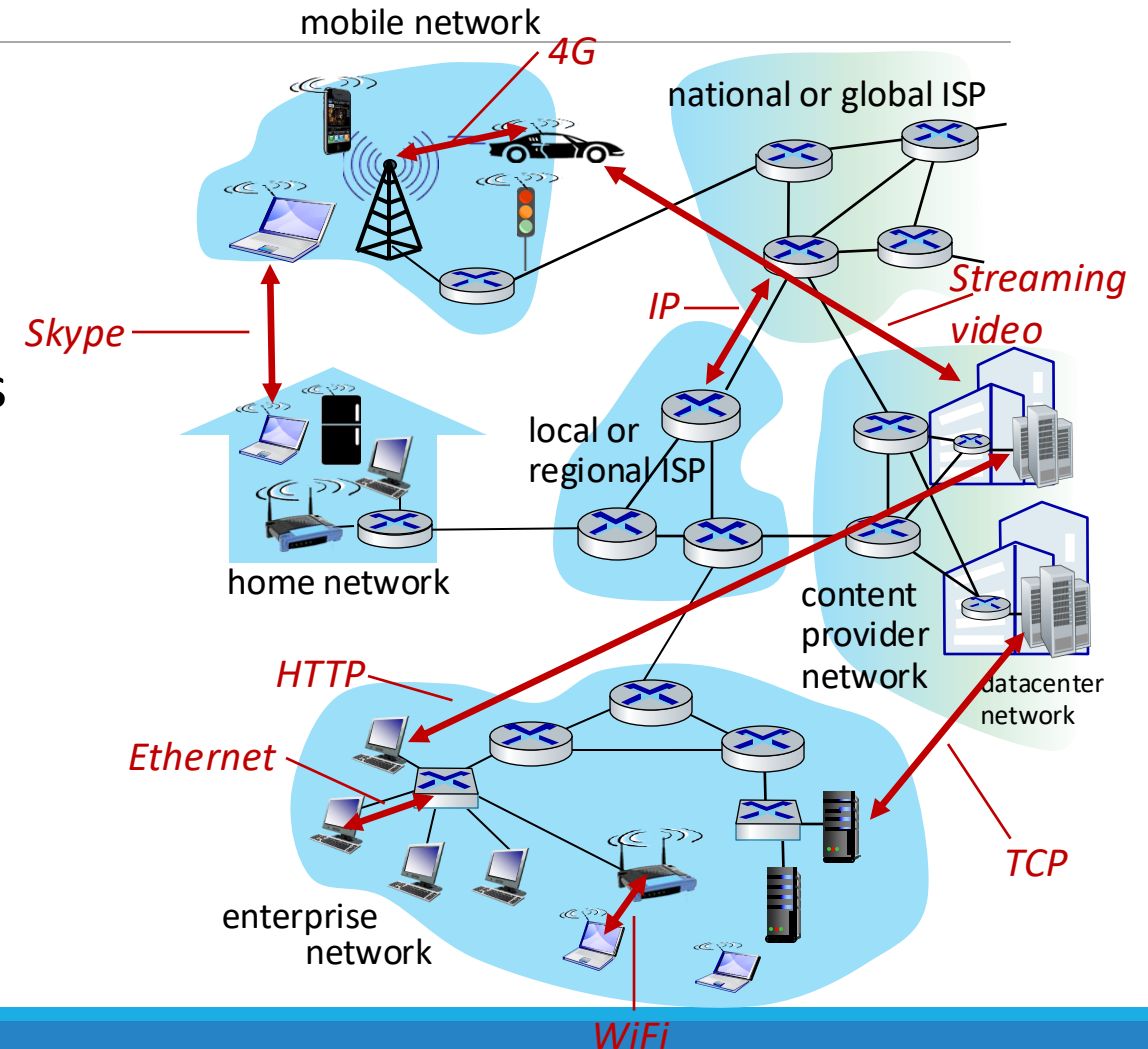
- Interconnected ISPs

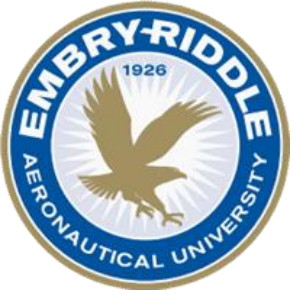
■ *protocols* are everywhere

- control sending, receiving of messages
- e.g., HTTP (Web), streaming video, Skype, TCP, UDP, IP, WiFi, 4G, Ethernet

■ *Internet standards*

- RFC: Request for Comments
- IETF: Internet Engineering Task Force

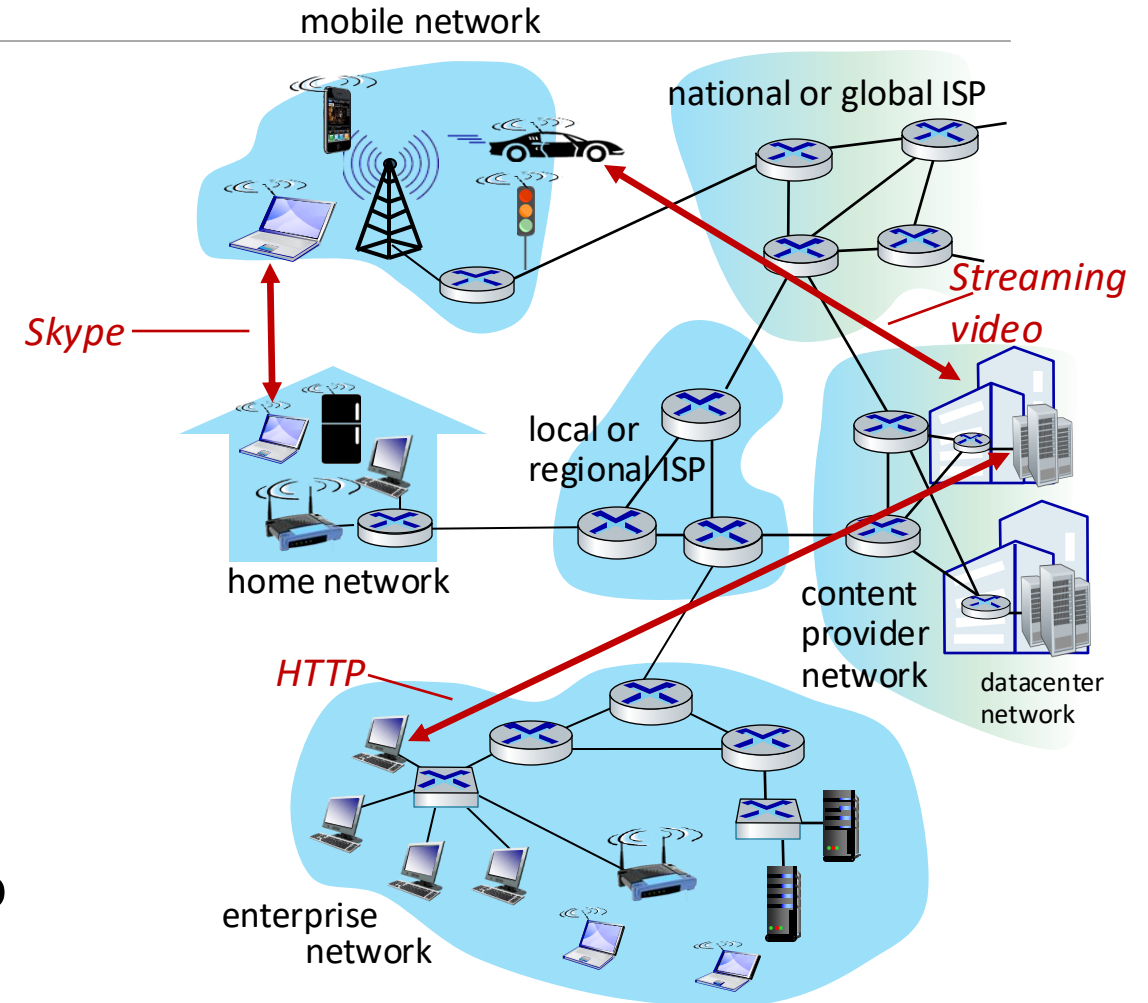


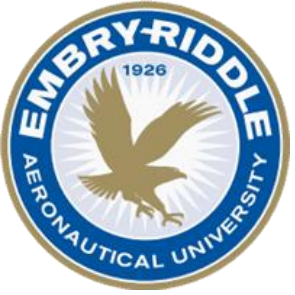


The Internet: a “services” view

Infrastructure that provides services to applications:

- Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides *programming interface* to distributed applications:
 - Different devices allow sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service





Roadmap

What *is* the Internet?

What ***is*** a protocol?

Network edge: hosts, access network,
physical media

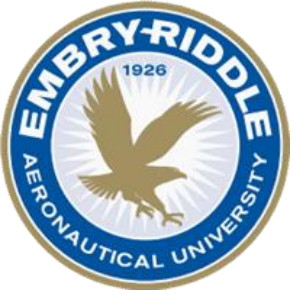
Network core: packet/circuit switching,
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History



What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

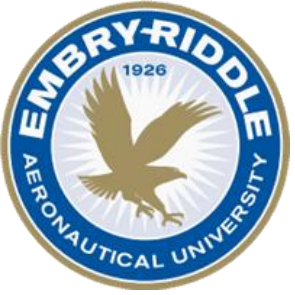
Rules for:

- ... specific messages sent
- ... specific actions taken
when message received,
or other events

Network protocols:

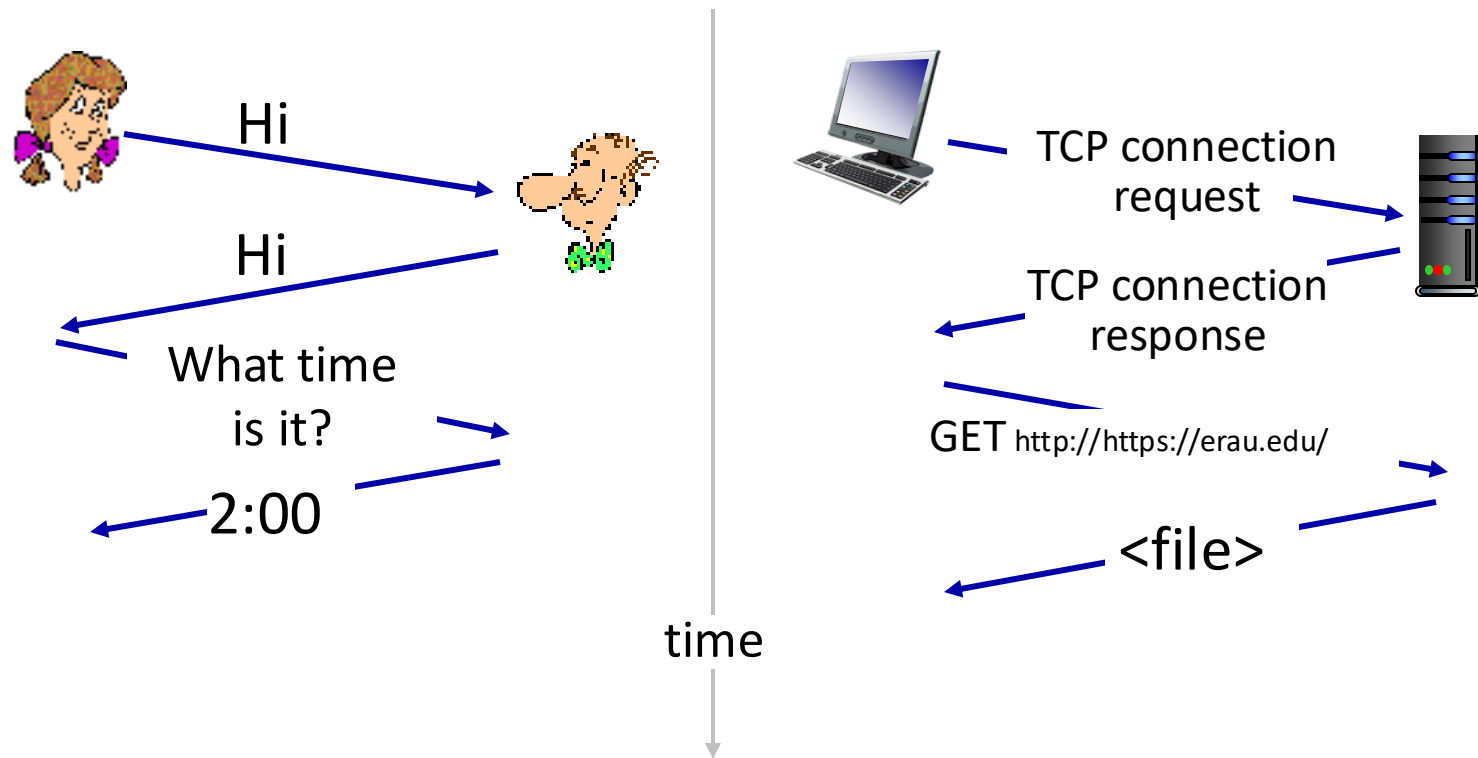
- computers (devices) rather than humans
- all communication activity in Internet
governed by protocols

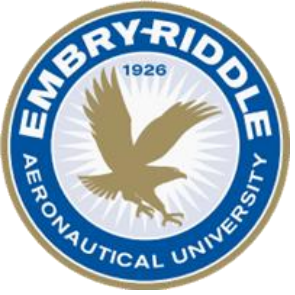
*Protocols define the **format, order** of
messages sent and received among
network entities, and **actions taken**
on message transmission, receipt*



What's a protocol?

A human protocol and a computer network protocol:





Roadmap

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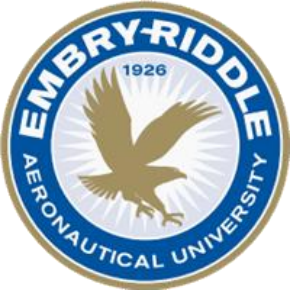
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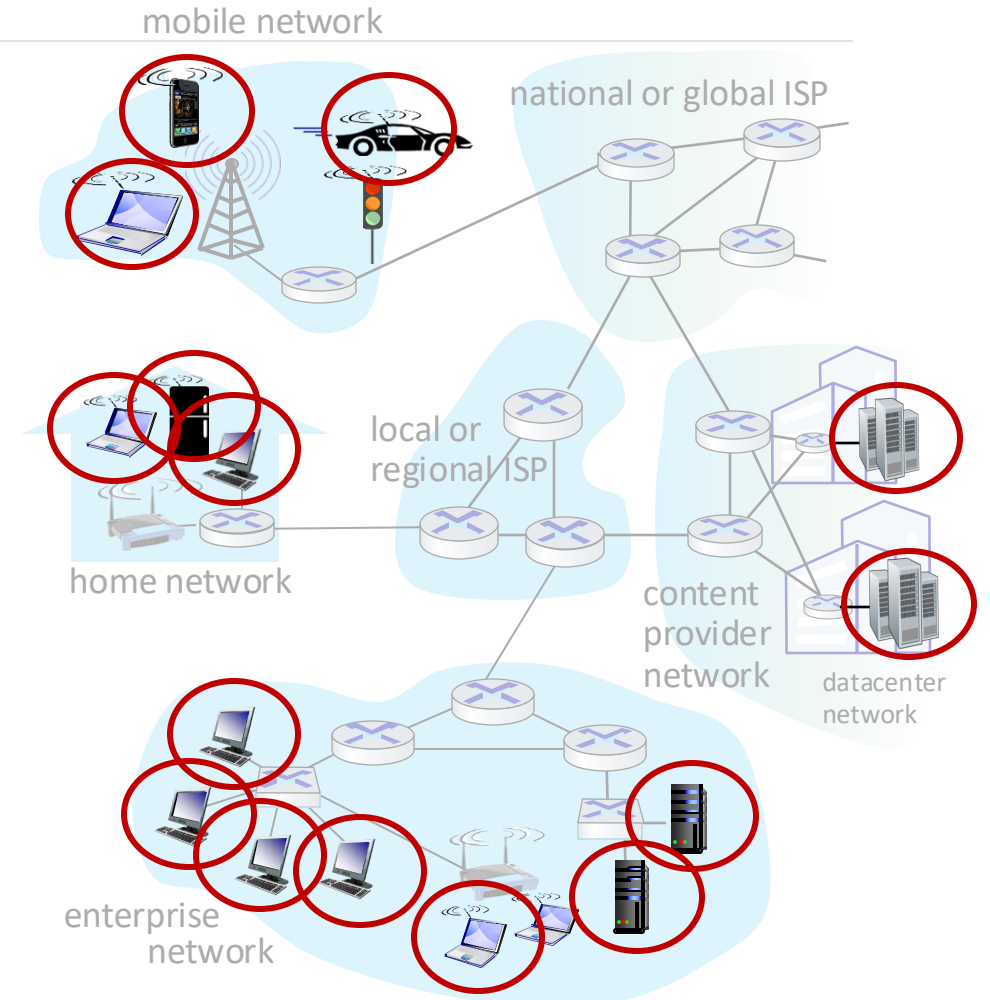


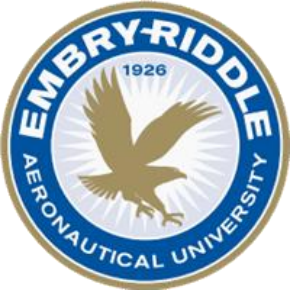
A closer look at Internet structure

Network edge:

hosts: clients and servers

servers often in data centers





A closer look at Internet structure

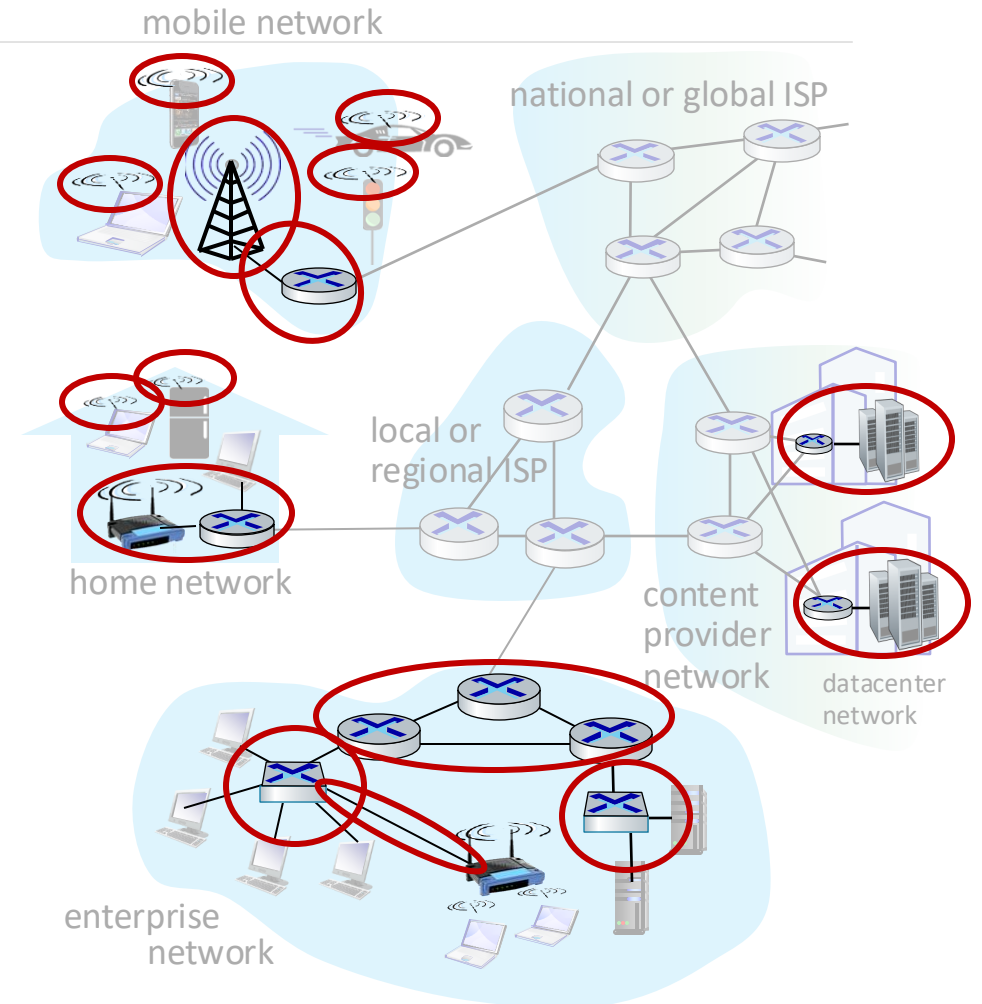
Network edge:

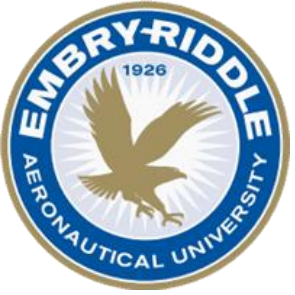
hosts: clients and servers

servers often in data centers

Access networks, physical media:

wired, wireless communication links





A closer look at Internet structure

Network edge:

hosts: clients and servers

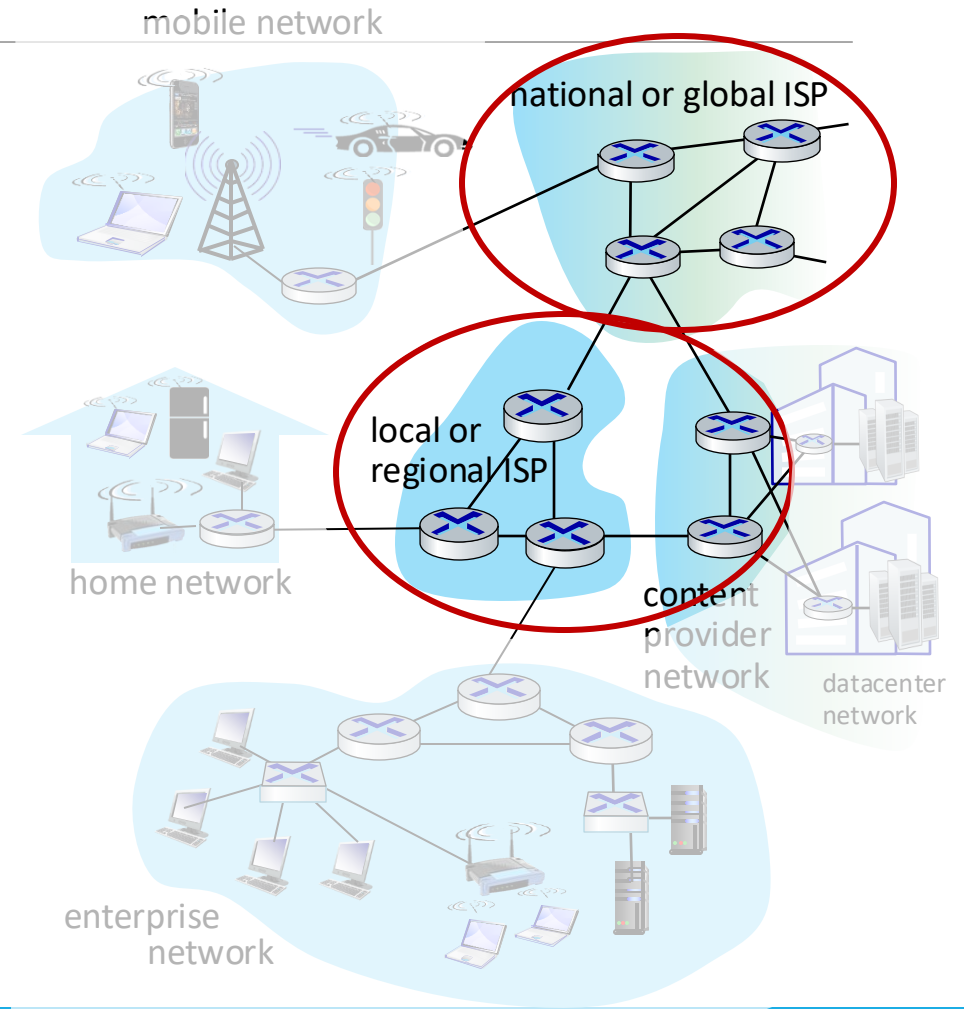
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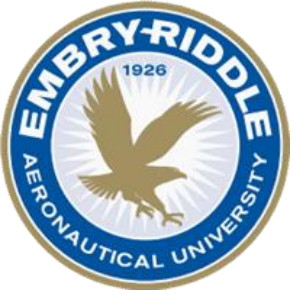
Access networks, physical media:

wired, wireless communication links

Network core:

- interconnected routers
- network of networks





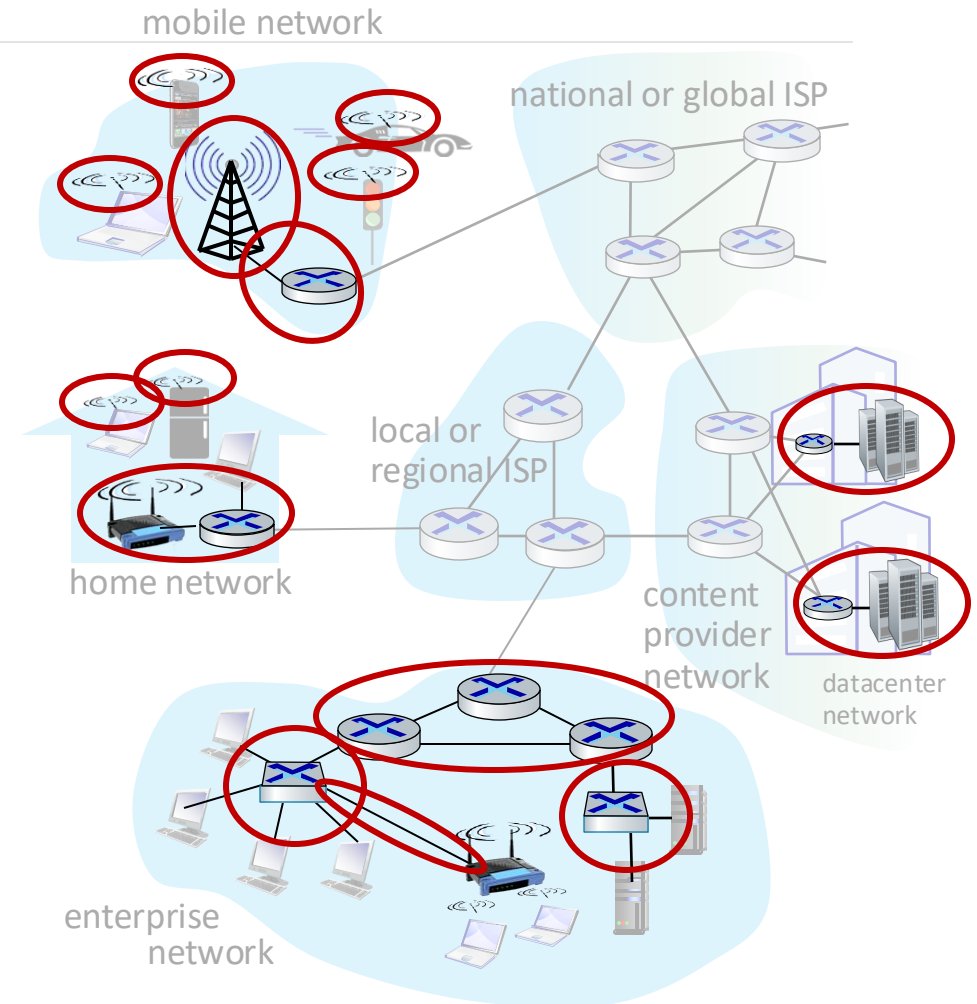
Access networks and physical media

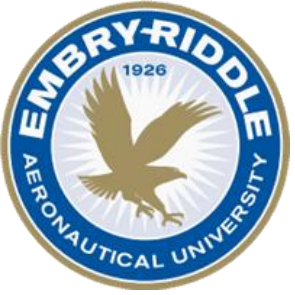
Q: How to connect end systems to edge router?

residential access nets

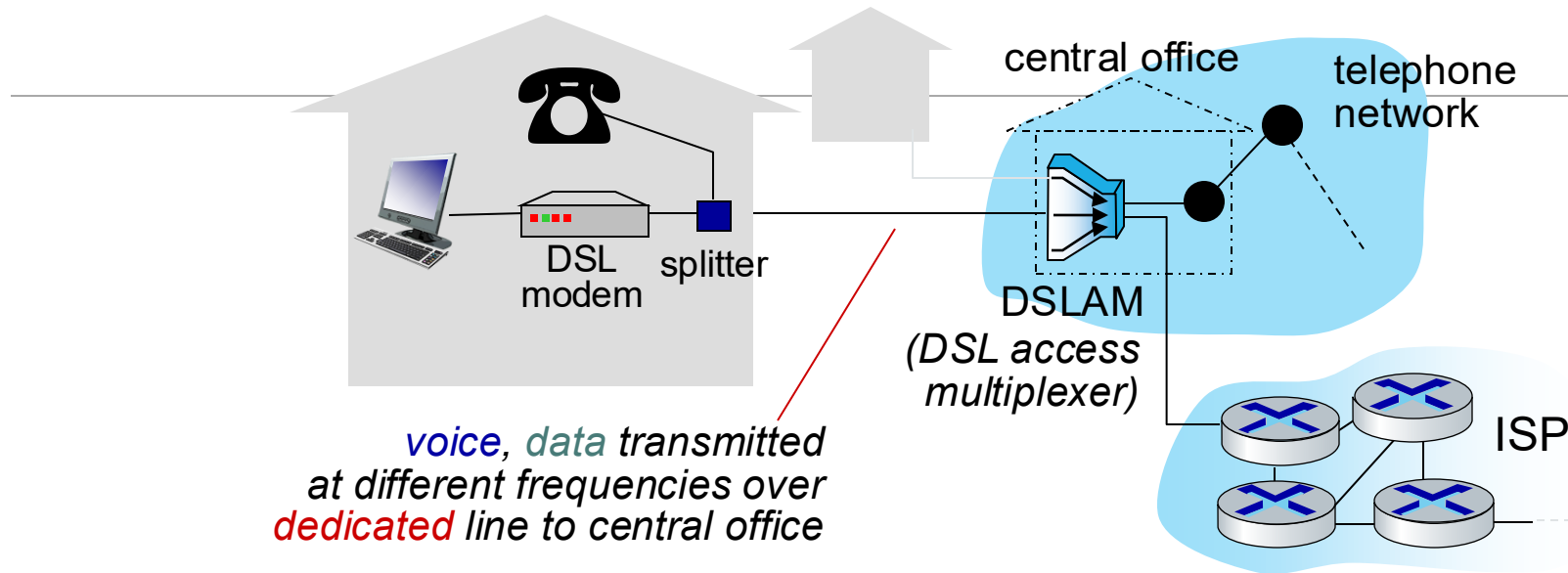
institutional access networks (school, company)

mobile access networks (WiFi, 4G/5G)



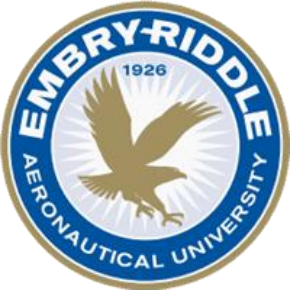


Access networks: digital subscriber line (DSL)

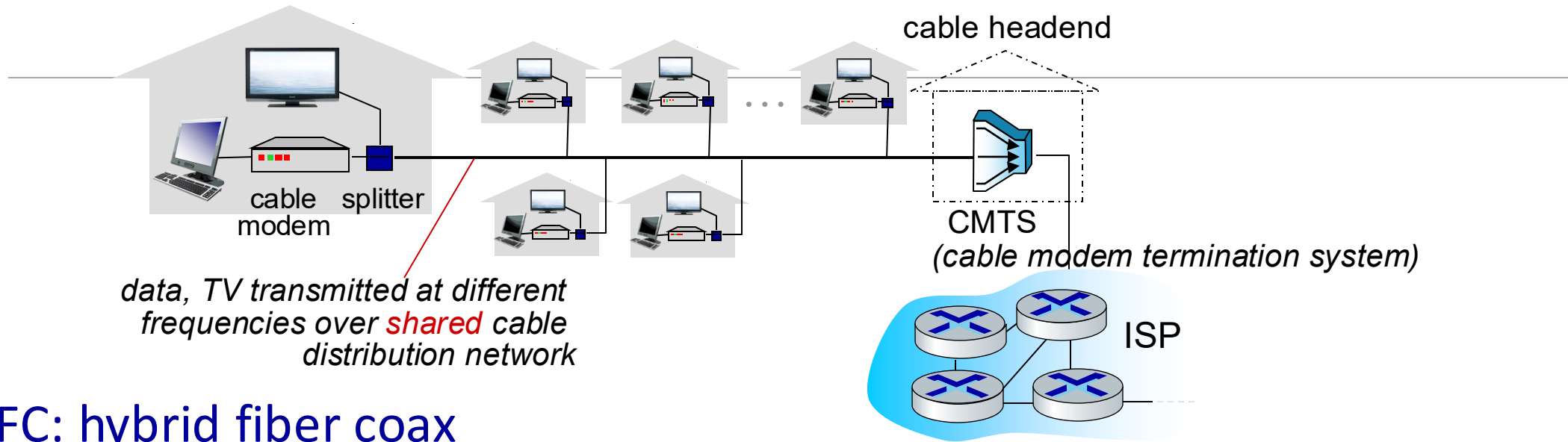


- use *existing* telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

DSL provides a dedicated access line from the customer to the telephone company's equipment. It is usually asymmetric: downloading is faster than uploading.



Access networks: cable-based access

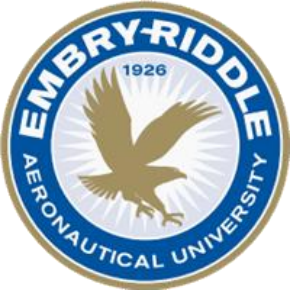


■ HFC: hybrid fiber coax

- asymmetric:
 - up to 40 Mbps – 1.2 Gbps downstream transmission rate
 - 30-100 Mbps upstream transmission rate

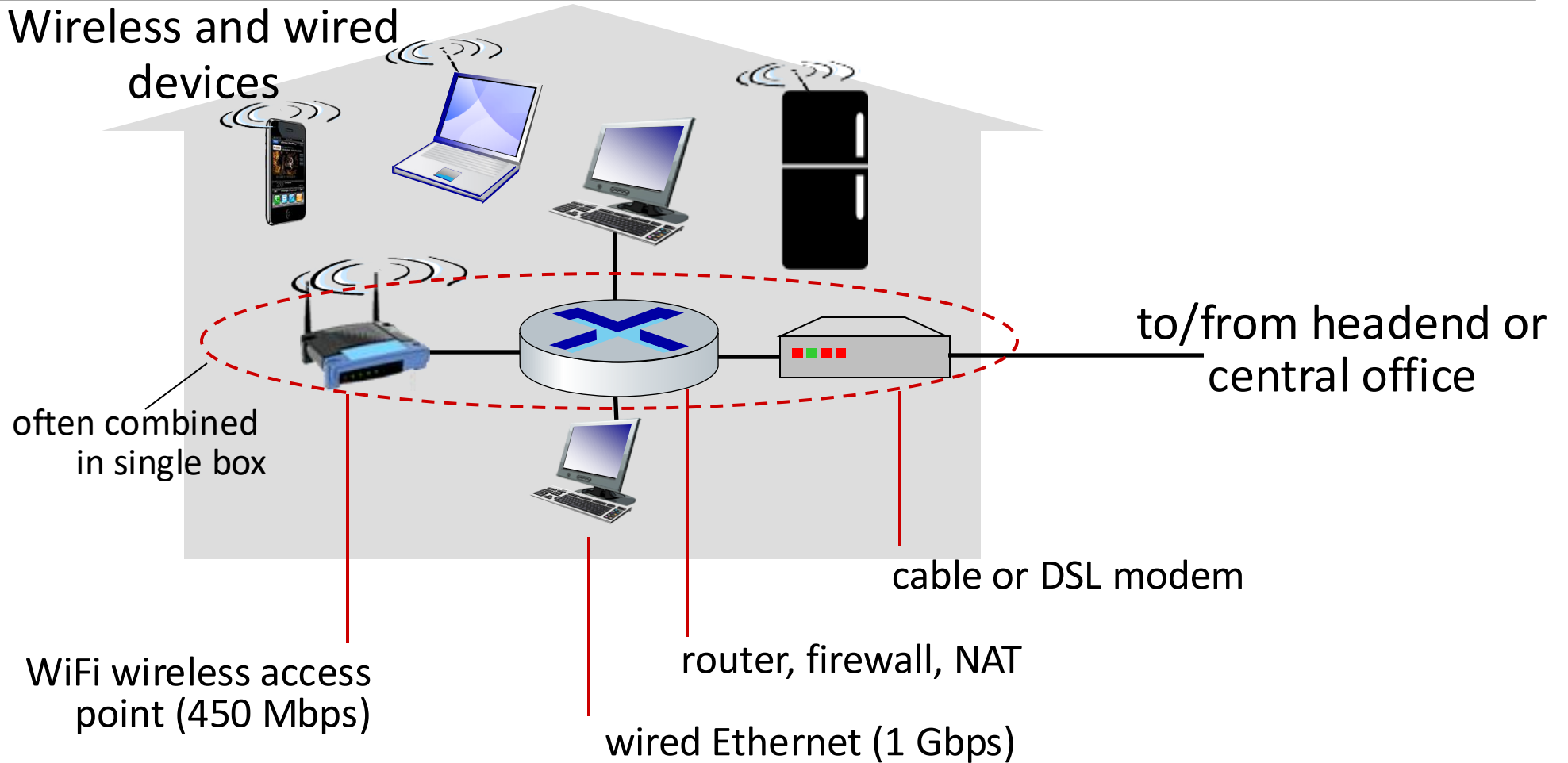
■ network of cable, fiber attaches homes to ISP router

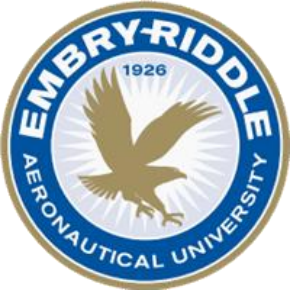
- homes *share access network* to cable headend



Access networks: home networks

Wireless and wired devices





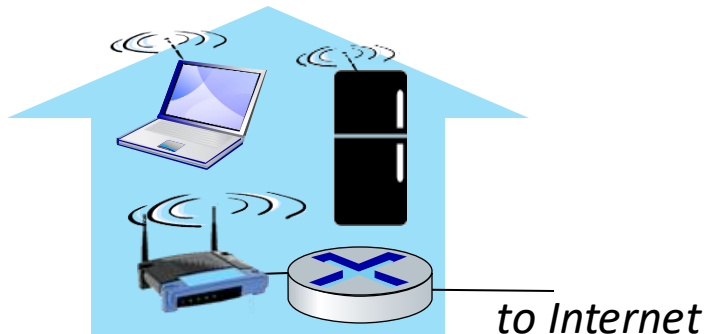
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

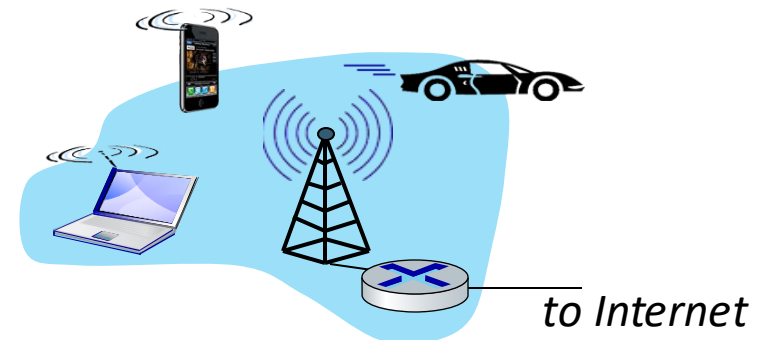
Wireless local area networks (WLANs)

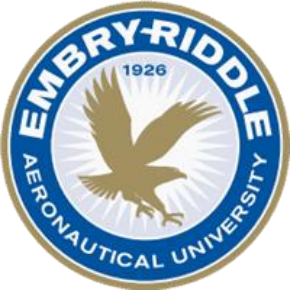
- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate



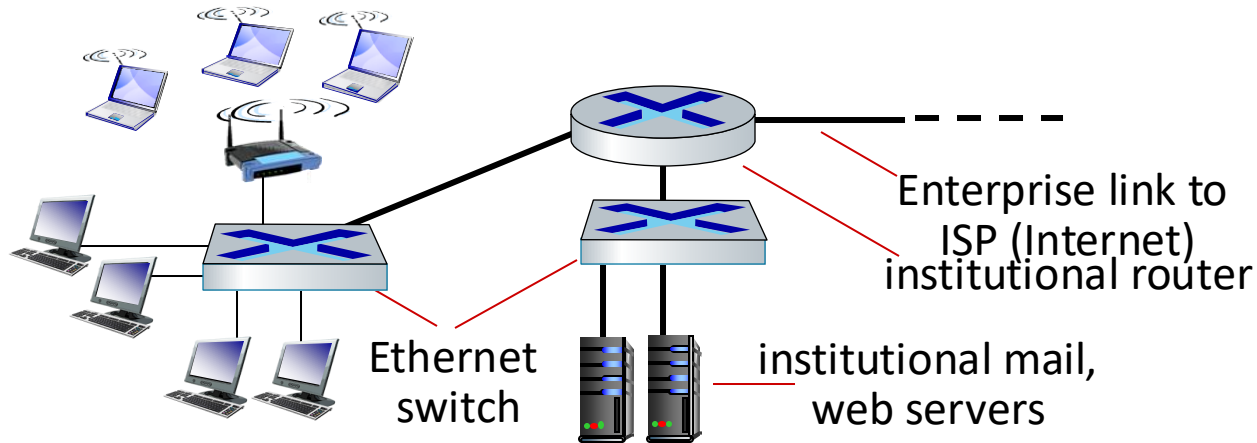
Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 5G cellular networks

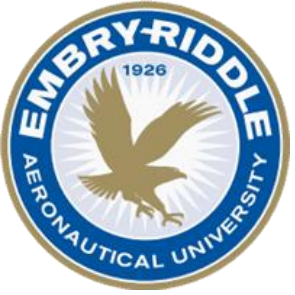




Access networks: enterprise networks



- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - WiFi: wireless access points at 11, 54, 450 Mbps

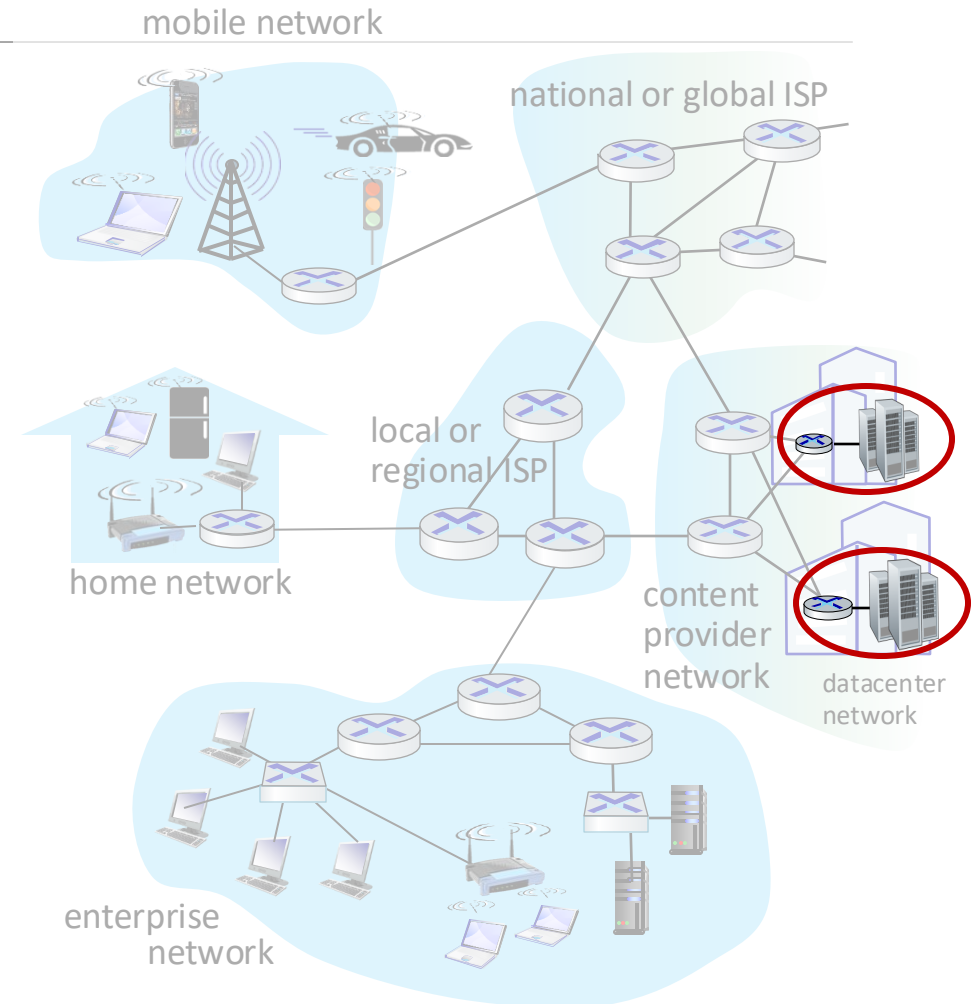


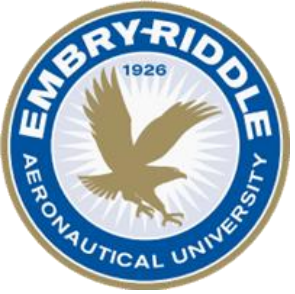
Access networks: data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)

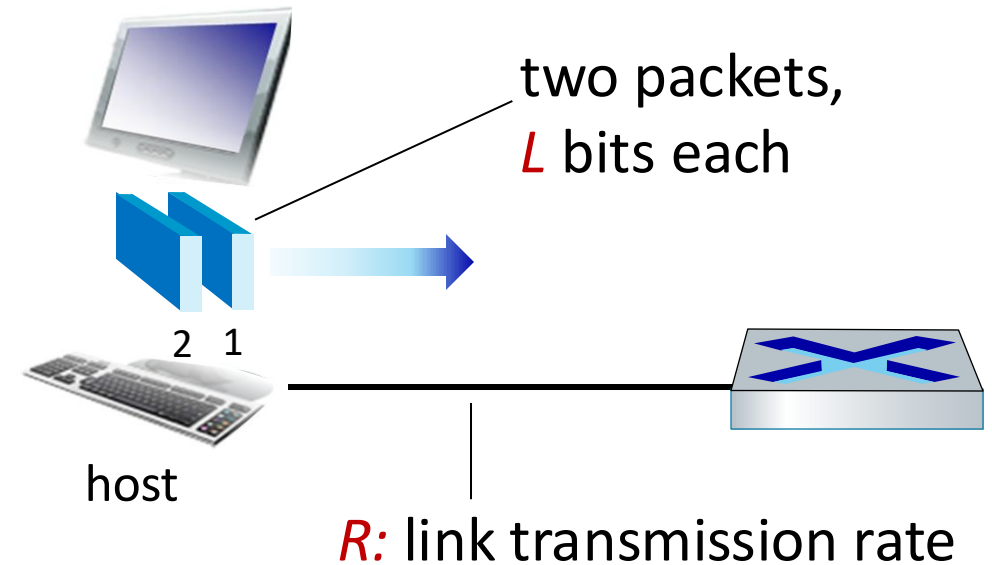




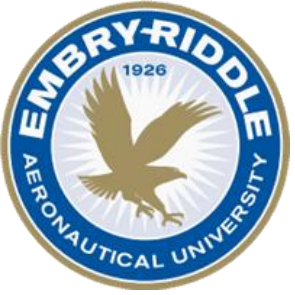
Host: sends *packets* of data

host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length L bits
- transmits packet into access network at *transmission rate R*
 - link transmission rate, aka link *capacity, aka link bandwidth*



$$\begin{array}{l} \text{packet} \\ \text{transmission} \\ \text{delay} \end{array} = \begin{array}{l} \text{time needed to} \\ \text{transmit } L\text{-bit} \\ \text{packet into link} \end{array} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

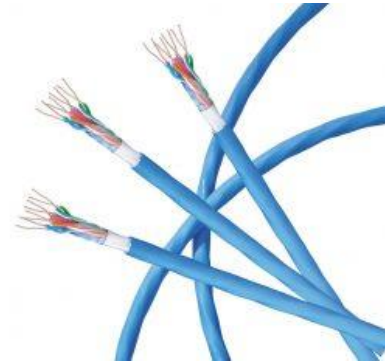


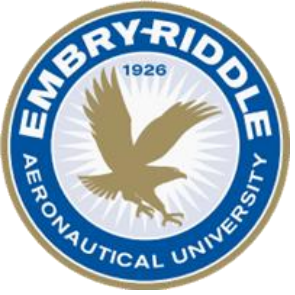
Links: physical media

- **bit**: propagates between transmitter/receiver pairs
- **physical link**: what lies between transmitter & receiver
- **guided media**:
 - signals propagate in solid media: copper, fiber, coax
- **unguided media**:
 - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet





Links: physical media

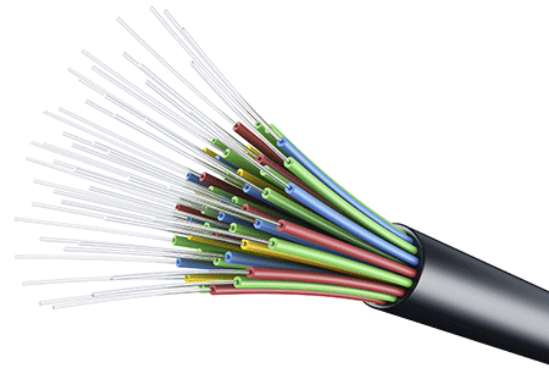
Coaxial cable:

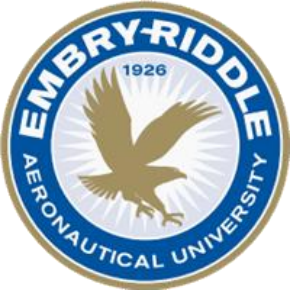
- two concentric copper conductors
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise





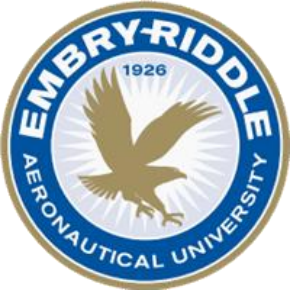
Links: physical media

Wireless radio

- signal carried in various “bands” in electromagnetic spectrum
- no physical “wire”
- broadcast, “half-duplex” (sender to receiver or receiver to sender)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - Interference/noise

Radio link types:

- **Wireless LAN (WiFi)**
 - 10-100’s Mbps; 10’s of meters
- **wide-area** (e.g., 4G cellular)
 - 10’s Mbps over ~10 Km
- **Bluetooth**: cable replacement
 - short distances, limited rates
- **terrestrial microwave**
 - point-to-point; 45 Mbps channels
- **satellite**
 - up to 45 Mbps per channel
 - 270 msec end-end delay



Roadmap

What *is* the Internet?

What *is* a protocol?

Network edge: hosts, access network,
physical media

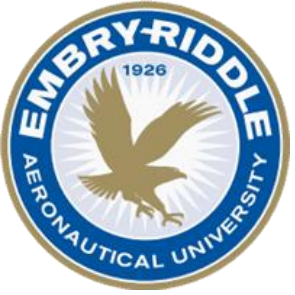
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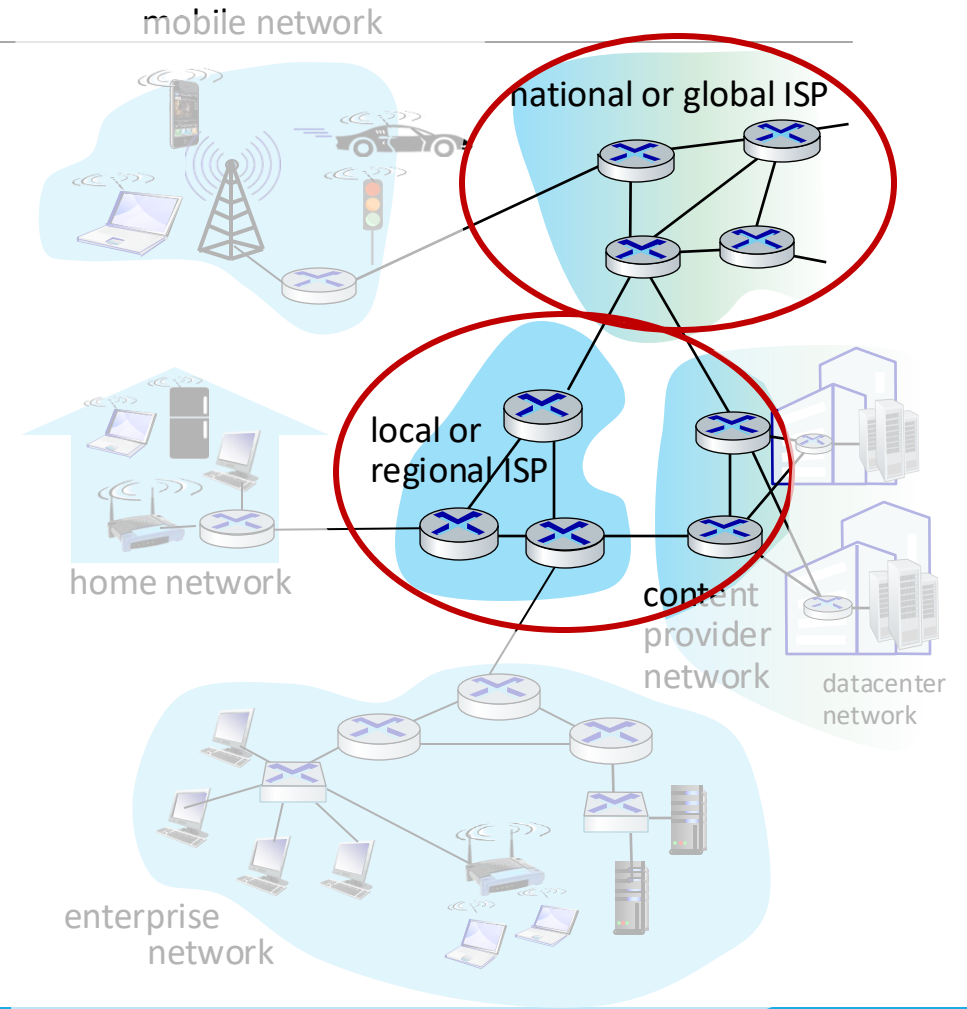


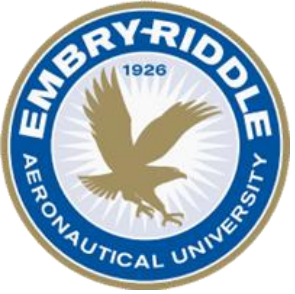
The network core

mesh of interconnected routers

packet-switching: hosts break application-layer messages into *packets*

- network **forwards** packets from one router to the next, across links on path from **source to destination**



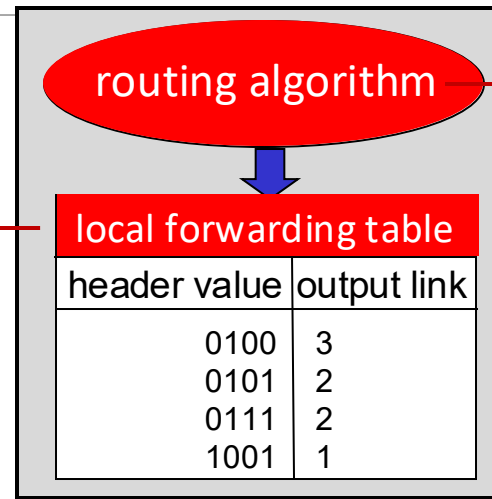


Two key network-core functions

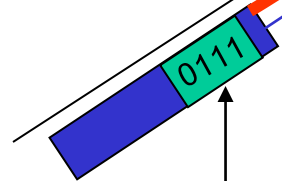
Forwarding:

aka “switching”

local action: move arriving packets from router’s input link to appropriate router output link

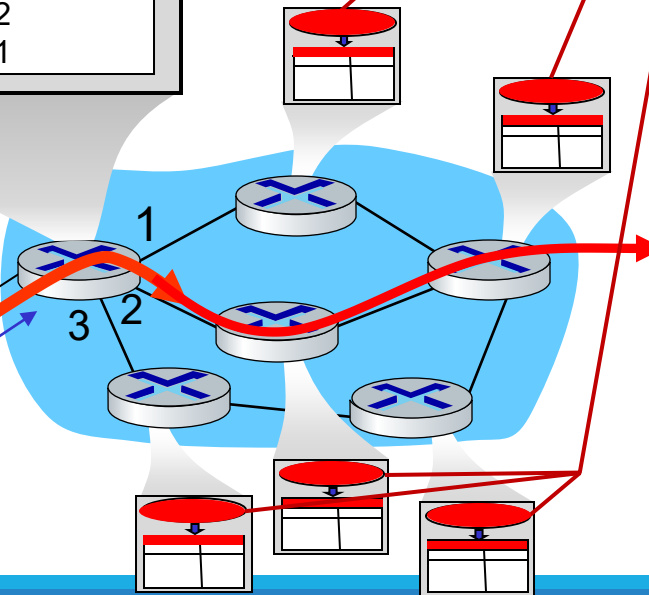


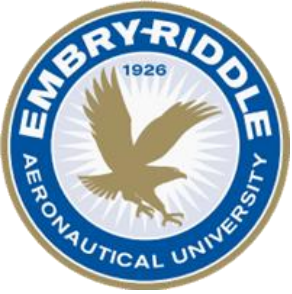
destination address in arriving packet's header

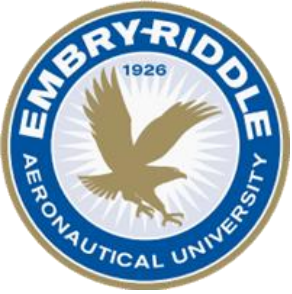


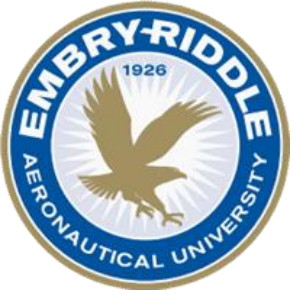
Routing:

- *global* action: determine source-destination paths taken by packets
- routing algorithms

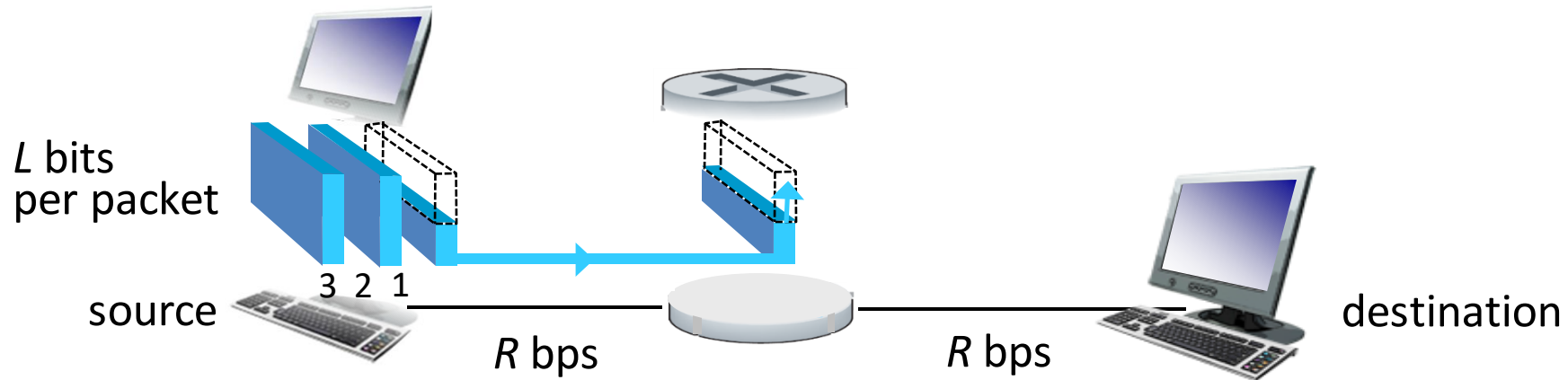








Packet-switching: store-and-forward

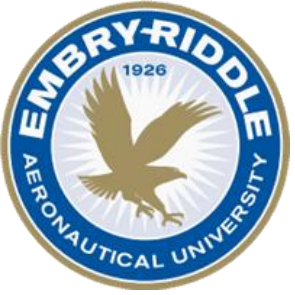


packet transmission delay: takes L/R seconds to transmit (push out) L -bit packet into link at R bps

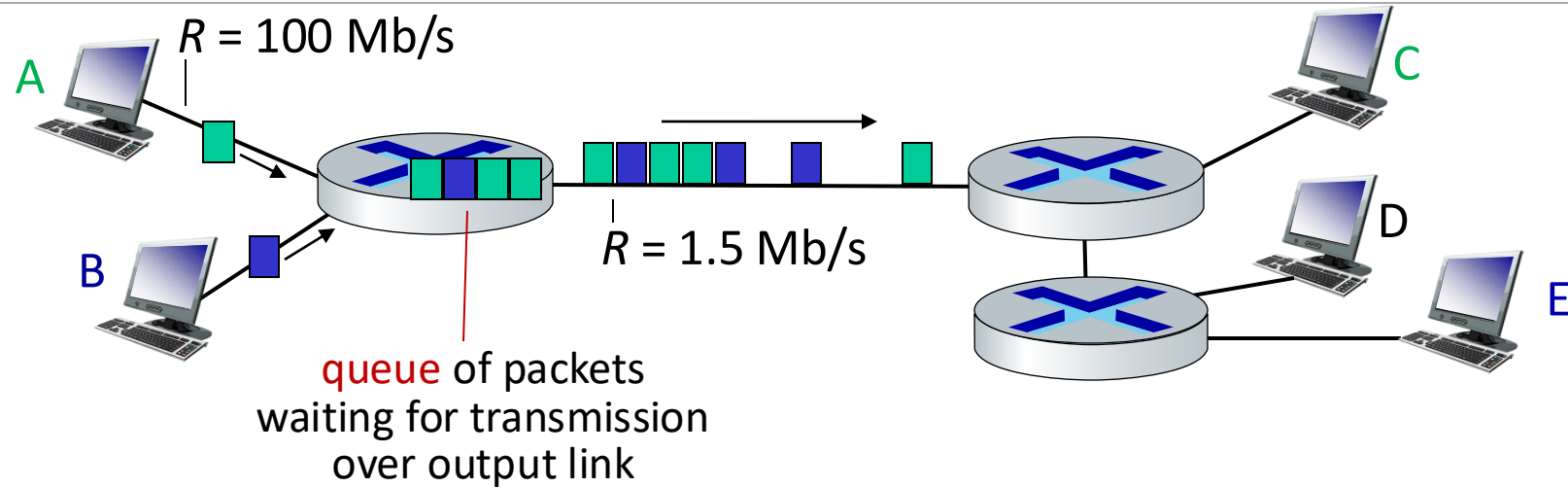
store and forward: entire packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ Mbps
- one-hop transmission delay = 0.1 msec

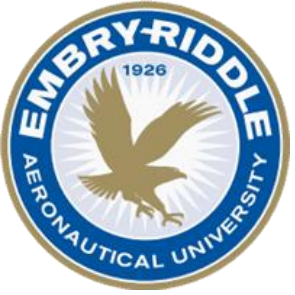


Packet-switching: queueing

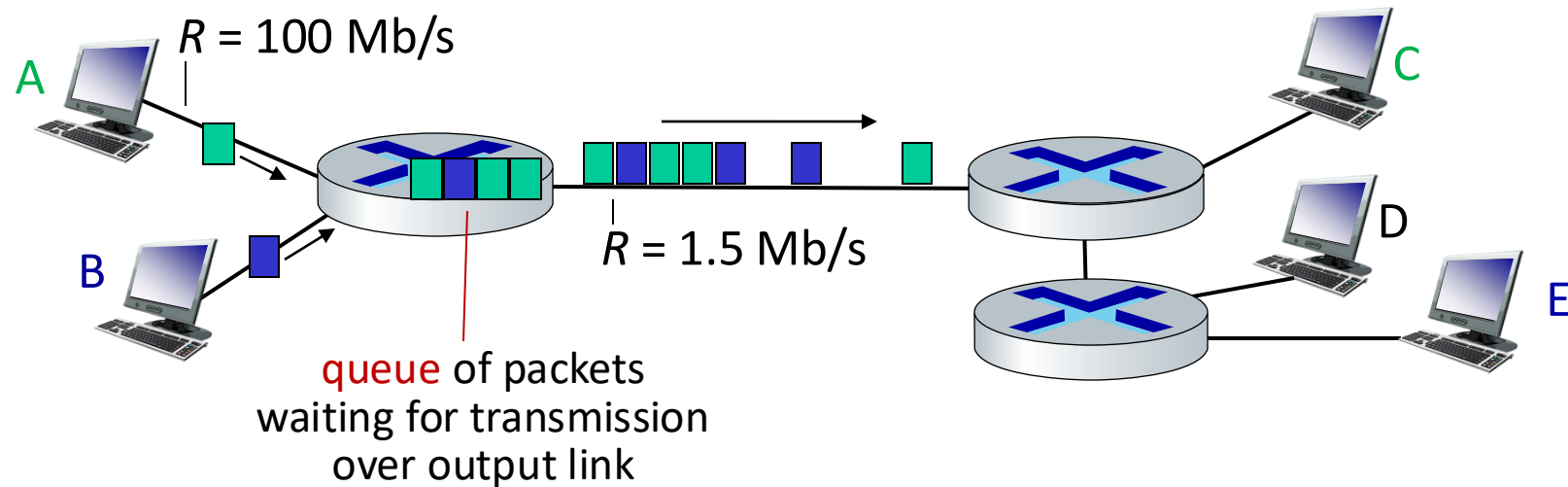


Queueing occurs when work arrives faster than it can be serviced:





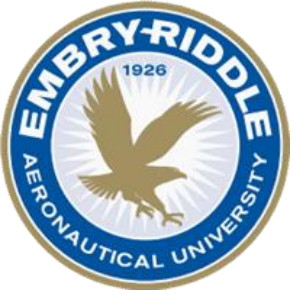
Packet-switching: queueing



Packet queueing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

packets will queue, waiting to be transmitted on output link

packets can be dropped (lost) if memory (buffer) in router fills up



Alternative to packet switching: circuit switching

end-end resources allocated to, reserved for “call”
between source and destination

in this diagram, each link has four circuits.

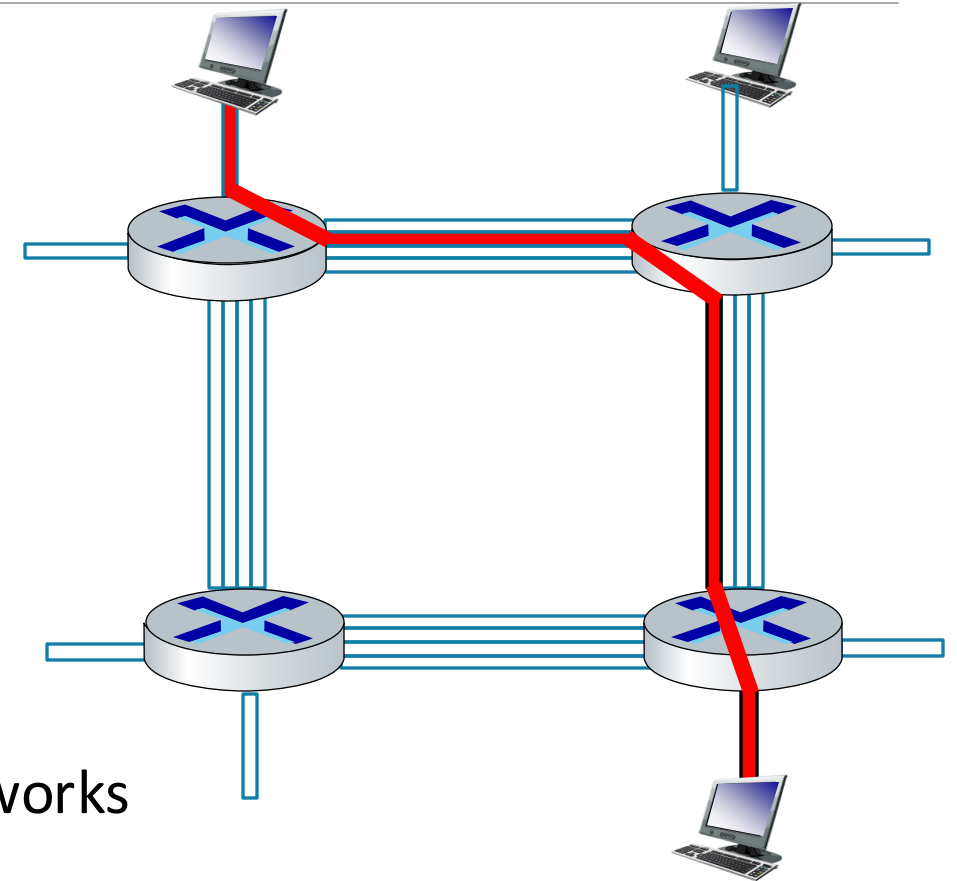
- call gets 2nd circuit in top link and 1st circuit in right link.

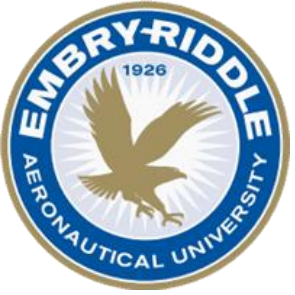
dedicated resources: no sharing

- circuit-like (guaranteed) performance

circuit segment idle if not used by call (no sharing)

- commonly used in traditional telephone networks



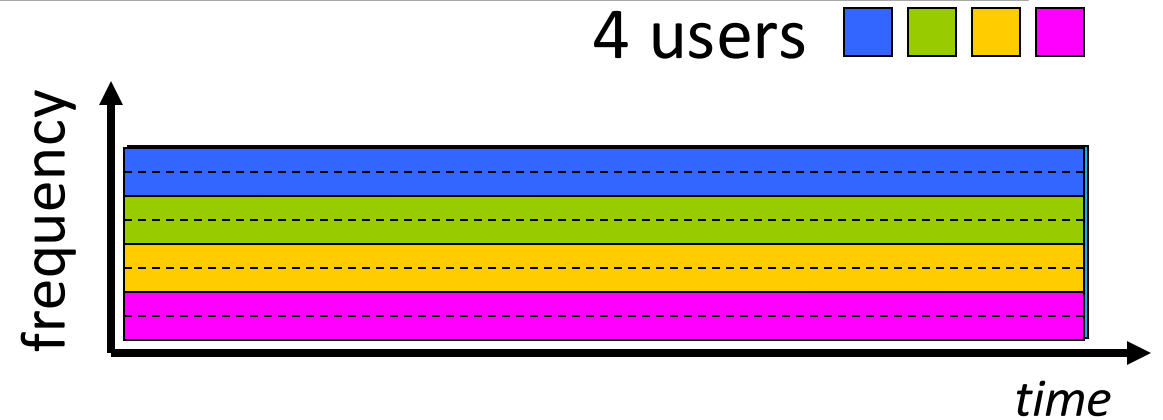


Circuit switching: FDM and TDM

Frequency Division Multiplexing (FDM)

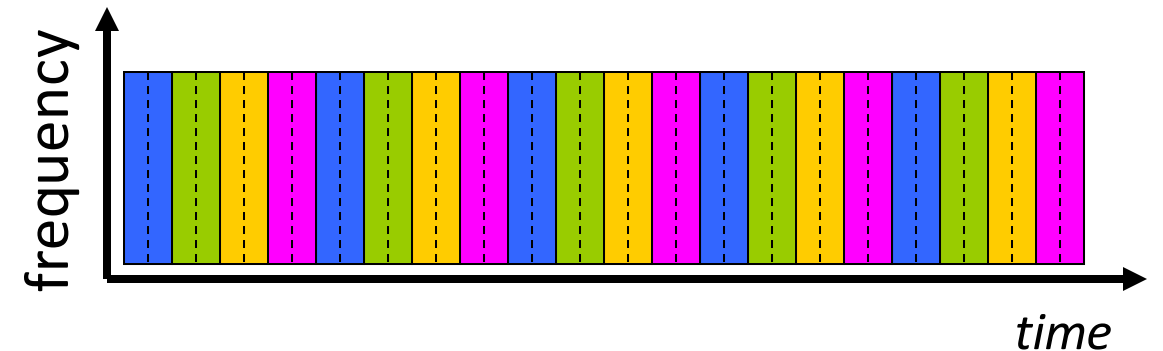
optical, electromagnetic frequencies divided into (narrow) frequency bands

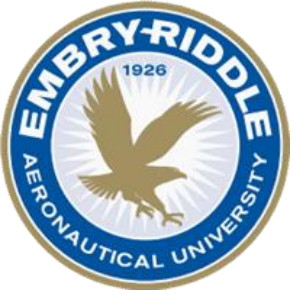
- Each call allocated its own band, can transmit at max rate of that narrow band



Time Division Multiplexing (TDM)

- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)

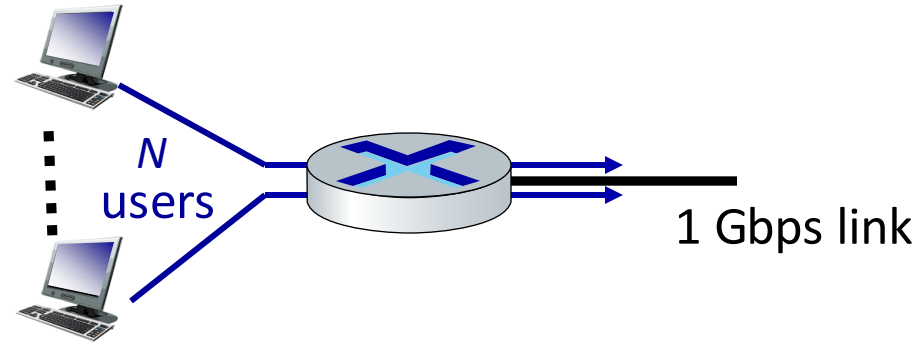




Packet switching versus circuit switching

Example:

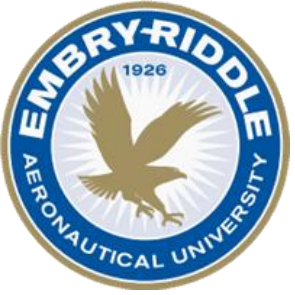
- 1 Gb/s link
- each user:
 - 100 Mb/s when “active”
 - active 10% of time



Q: how many users can use this network under circuit-switching and packet switching?

- *circuit-switching*: 10 users
- *packet switching - queueing*: with 35 users, probability > 10 active at same time is less than .0004 *

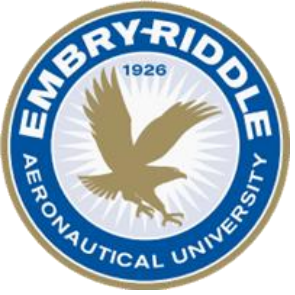
[Calculator](#)



Packet switching versus circuit switching

Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
 - resource sharing
 - simpler, no call setup
- **excessive congestion possible:** packet delay and loss due to buffer overflow
 - protocols needed for reliable data transfer, congestion control



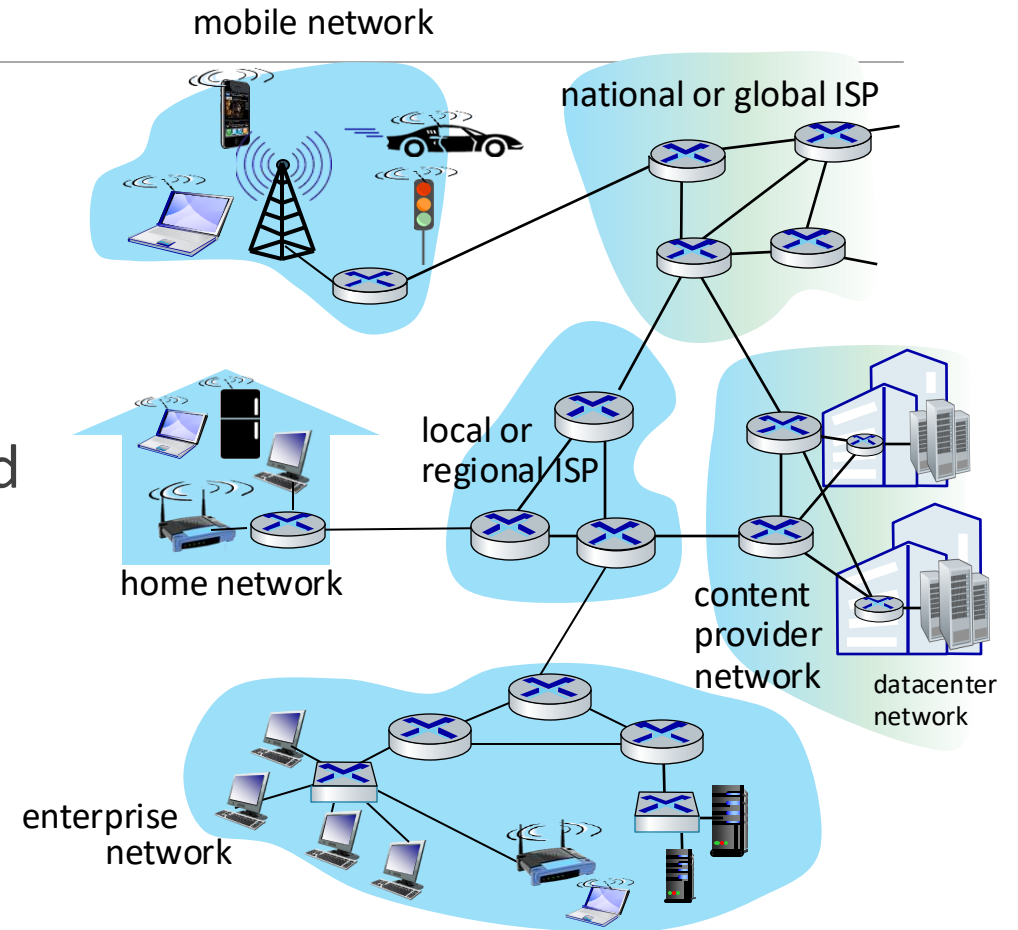
Internet structure: a “network of networks”

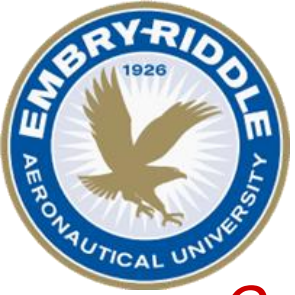
hosts connect to Internet via **access**
Internet Service Providers (ISPs)

access ISPs in turn must be
interconnected

- so that *any* two hosts (*anywhere!*) can send packets to each other

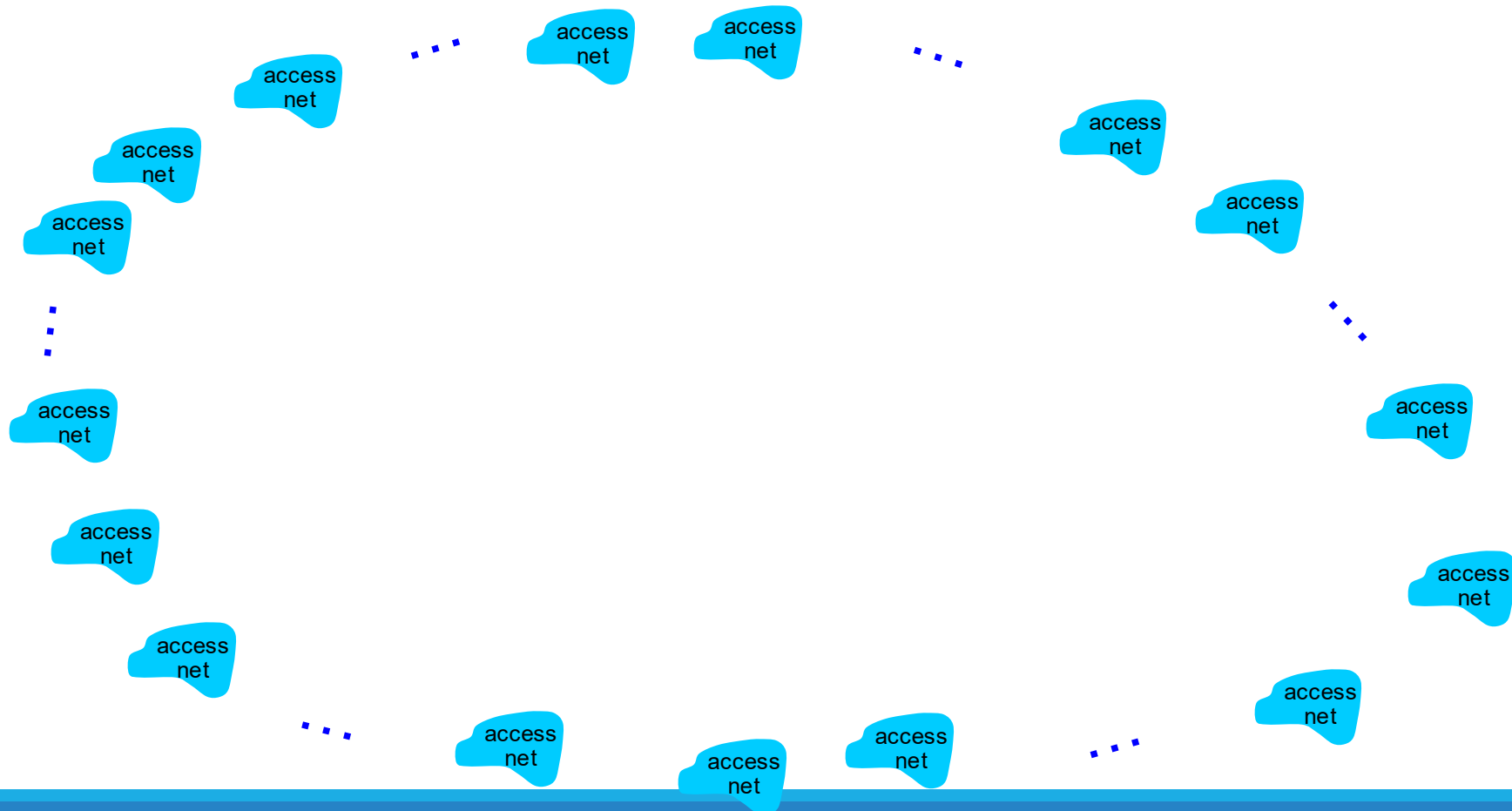
resulting network of networks is very
complex

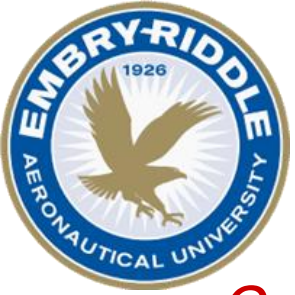




Internet structure: a “network of networks”

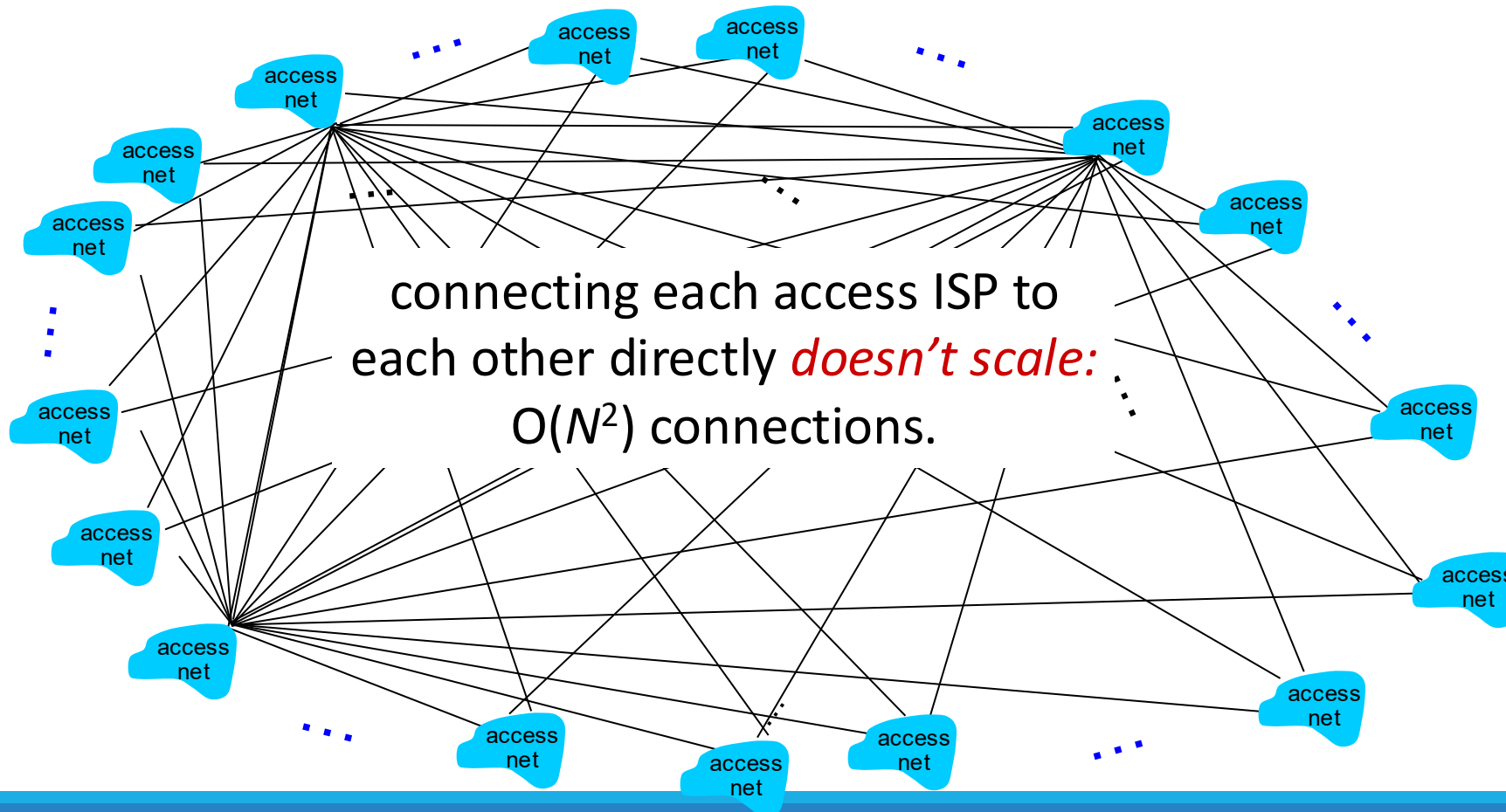
Question: given *millions* of access ISPs, how to connect them together?

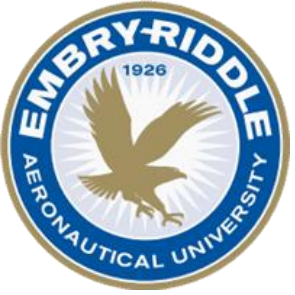




Internet structure: a “network of networks”

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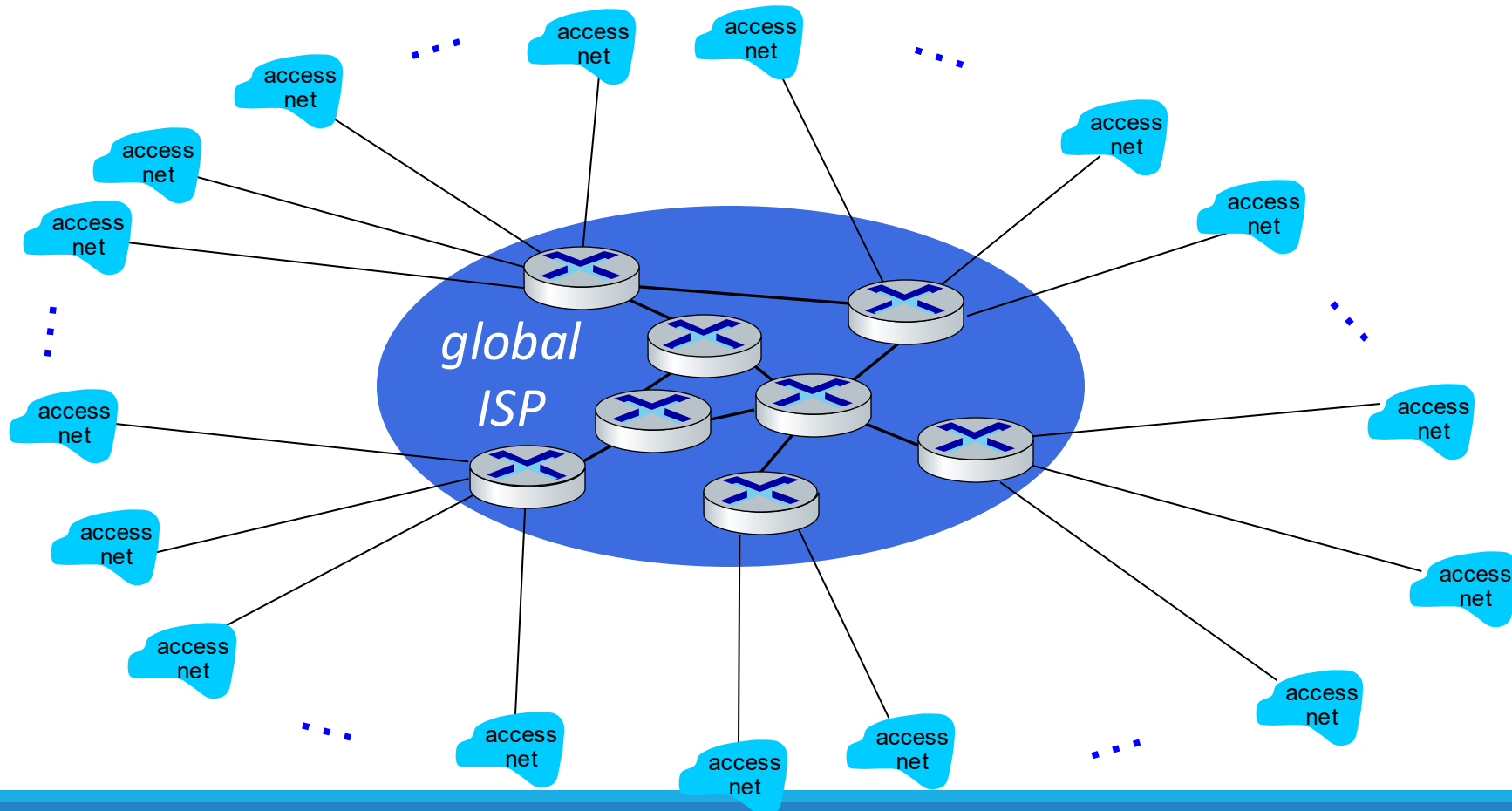


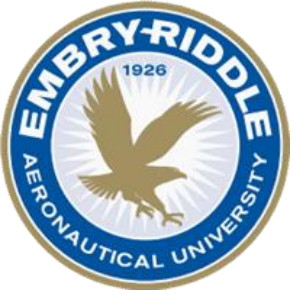


Internet structure: a “network of networks”

Option: connect each access ISP to one global transit ISP?

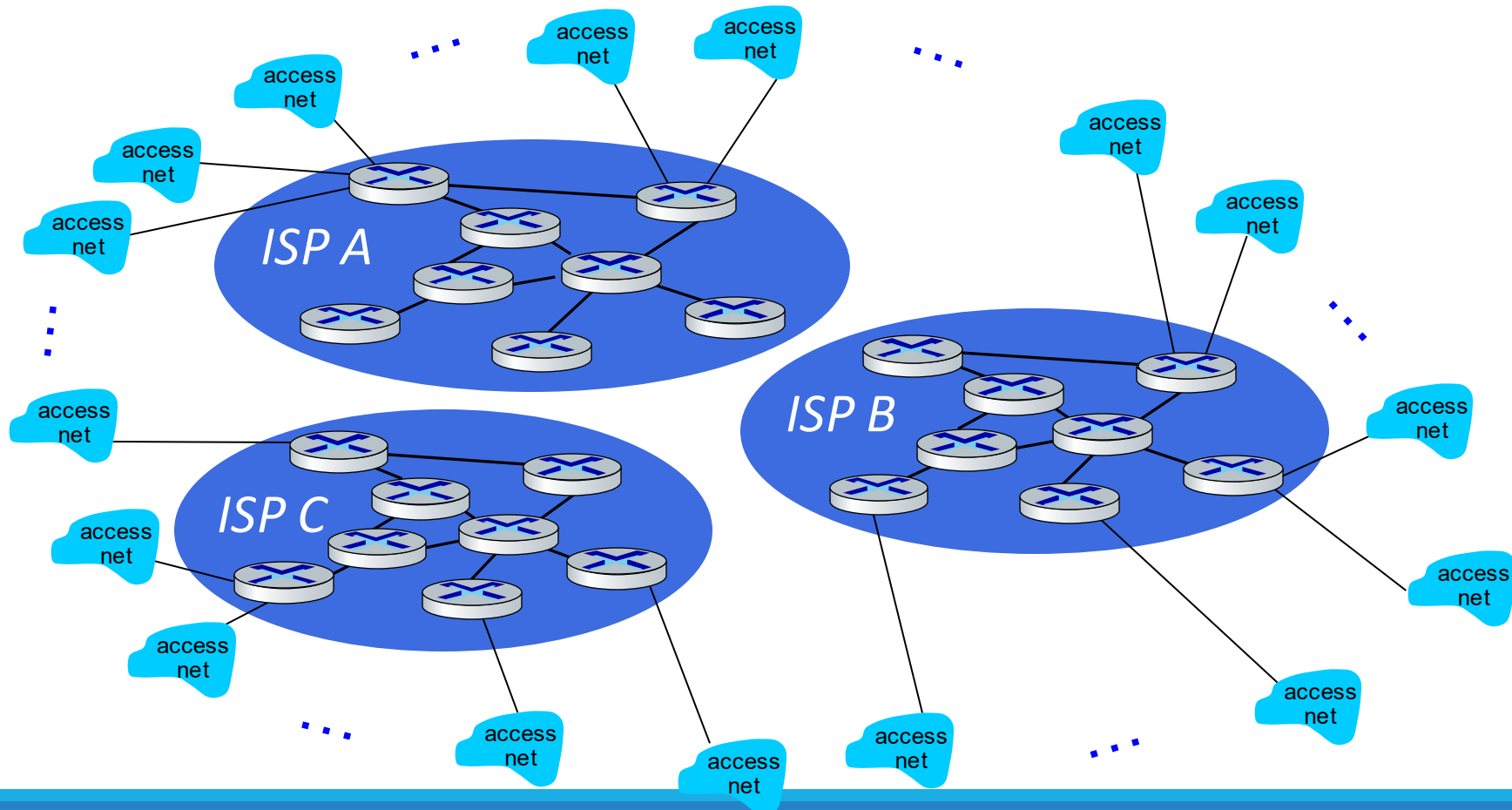
Customer and provider ISPs have economic agreement.

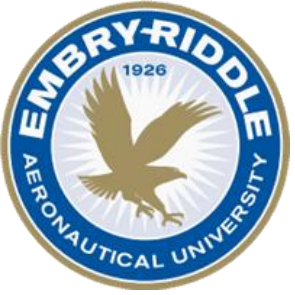




Internet structure: a “network of networks”

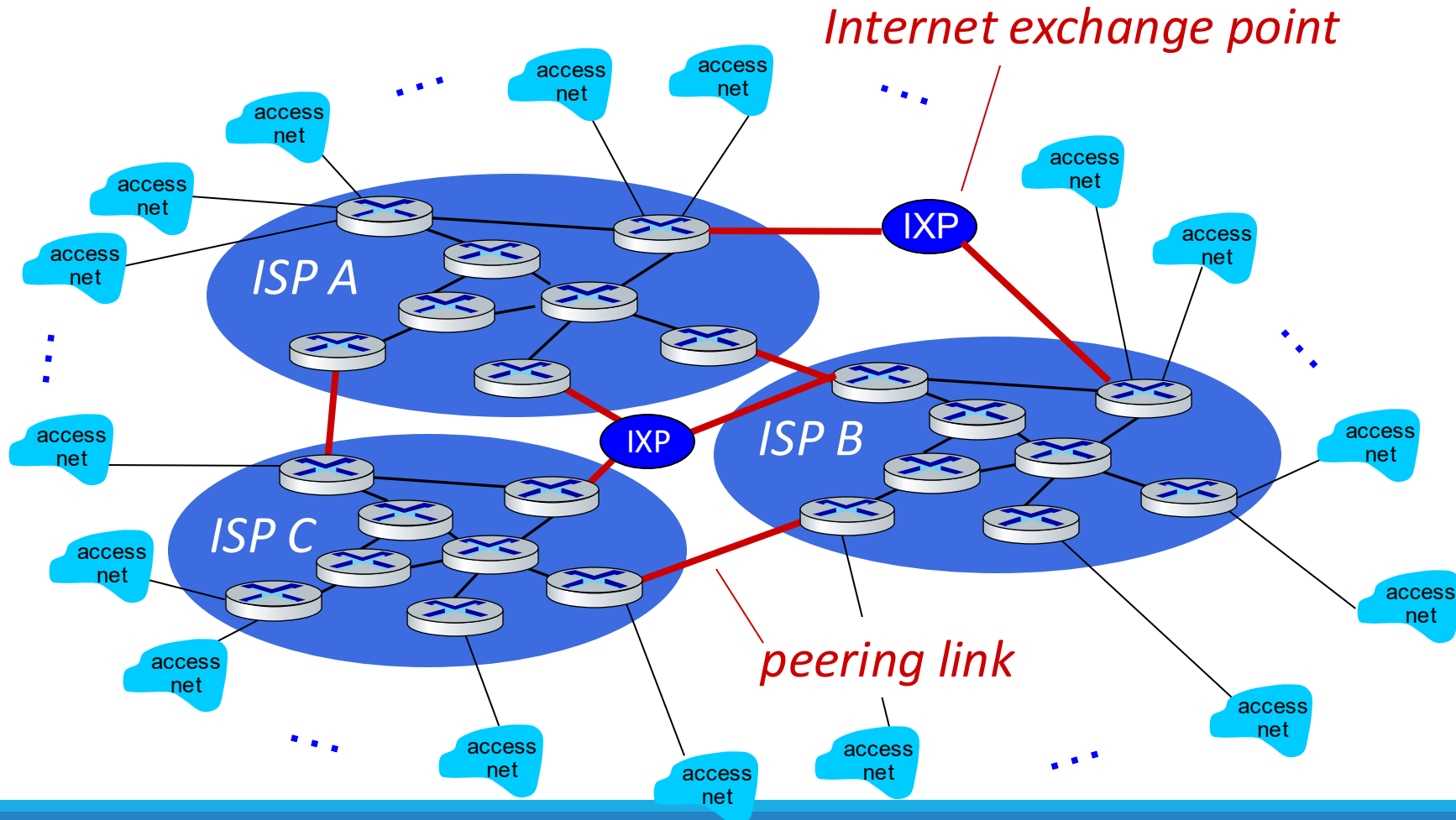
But if one global ISP is viable business, there will be competitors

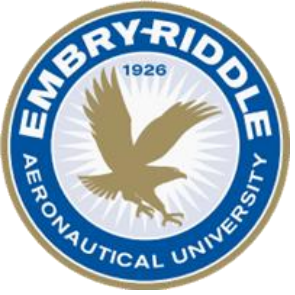




Internet structure: a “network of networks”

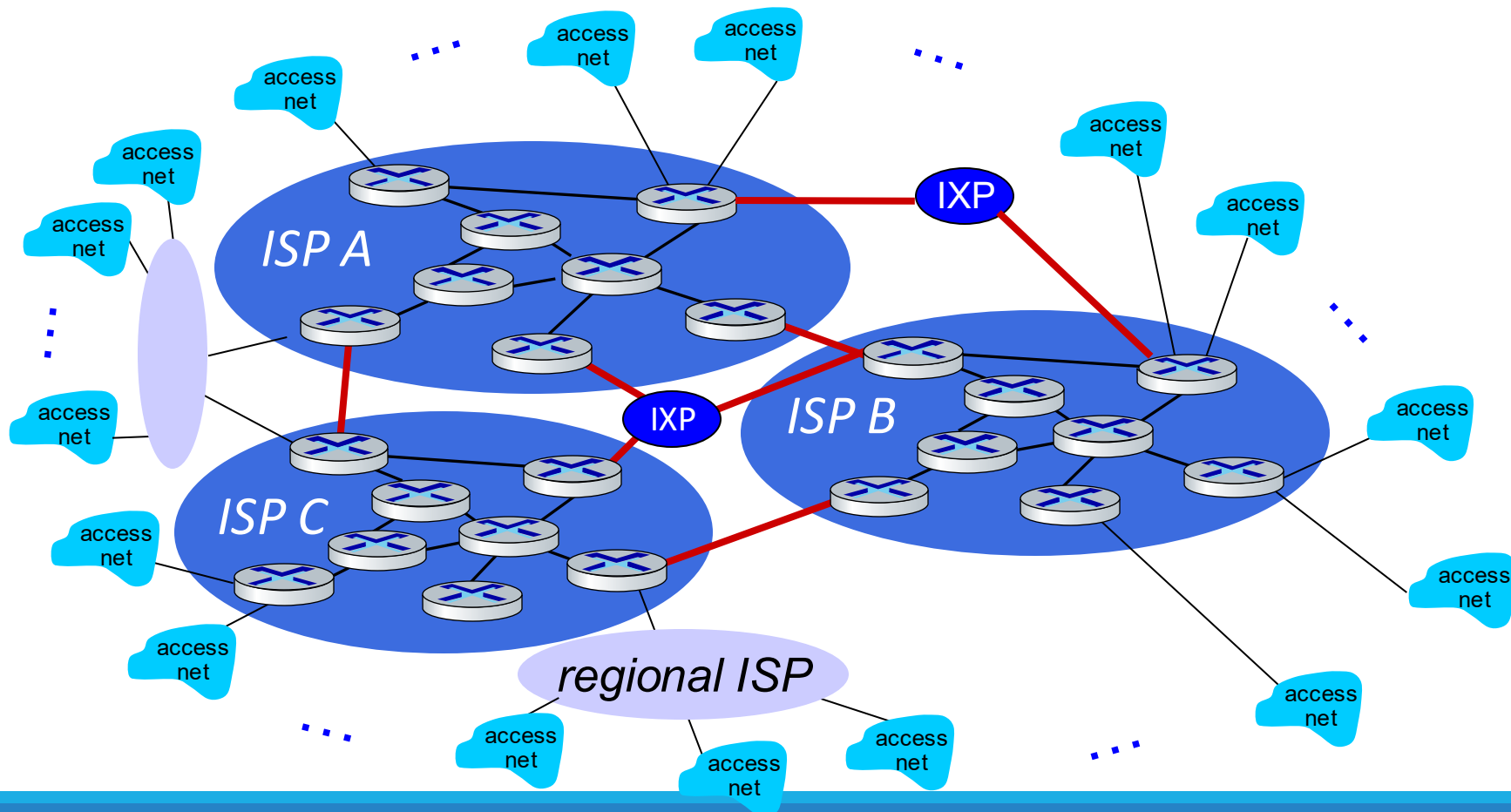
But if one global ISP is viable business, there will be competitors who will want to be connected

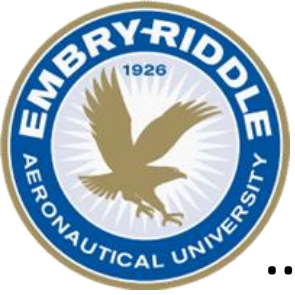




Internet structure: a “network of networks”

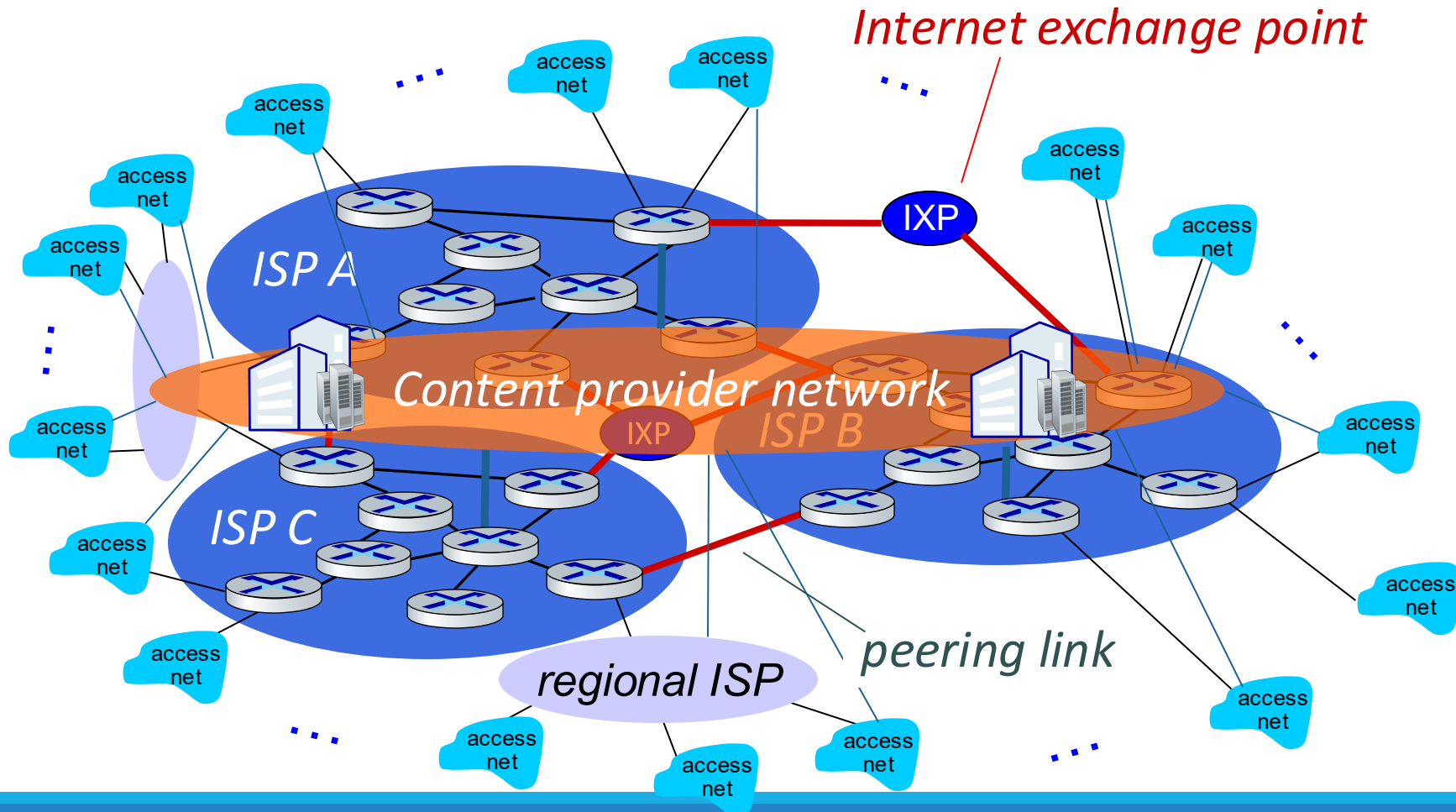
... and regional networks may arise to connect access nets to ISPs

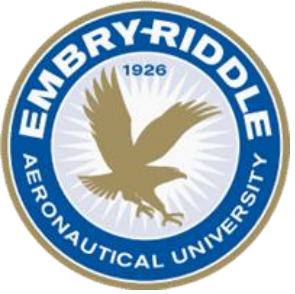




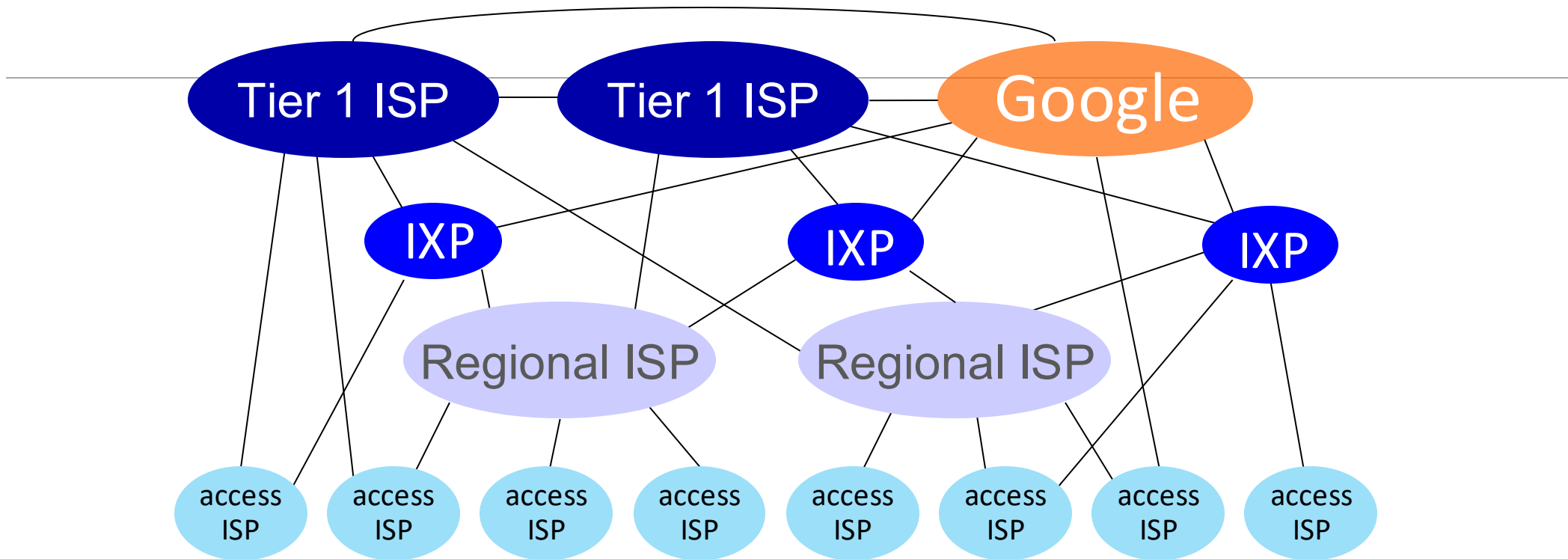
Internet structure: a “network of networks”

... and content provider networks (e.g., Google, Microsoft...) may run their own network, to bring services, content close to end users



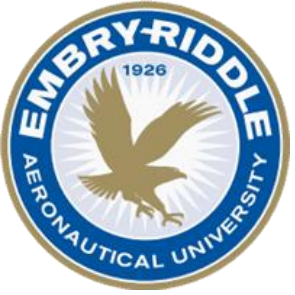


Internet structure: a “network of networks”



At “center”: small # of well-connected large networks

- **“tier-1” commercial ISPs** (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **content provider networks** (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs



Roadmap

What *is* the Internet?

What *is* a protocol?

Network edge: hosts, access network,
physical media

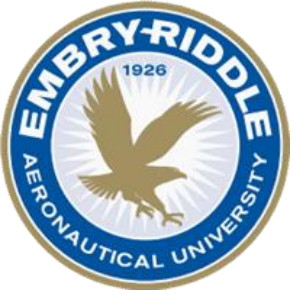
Network core: packet/circuit switching,
internet structure

Performance: delay, loss, throughput

Security

Protocol layers, service models

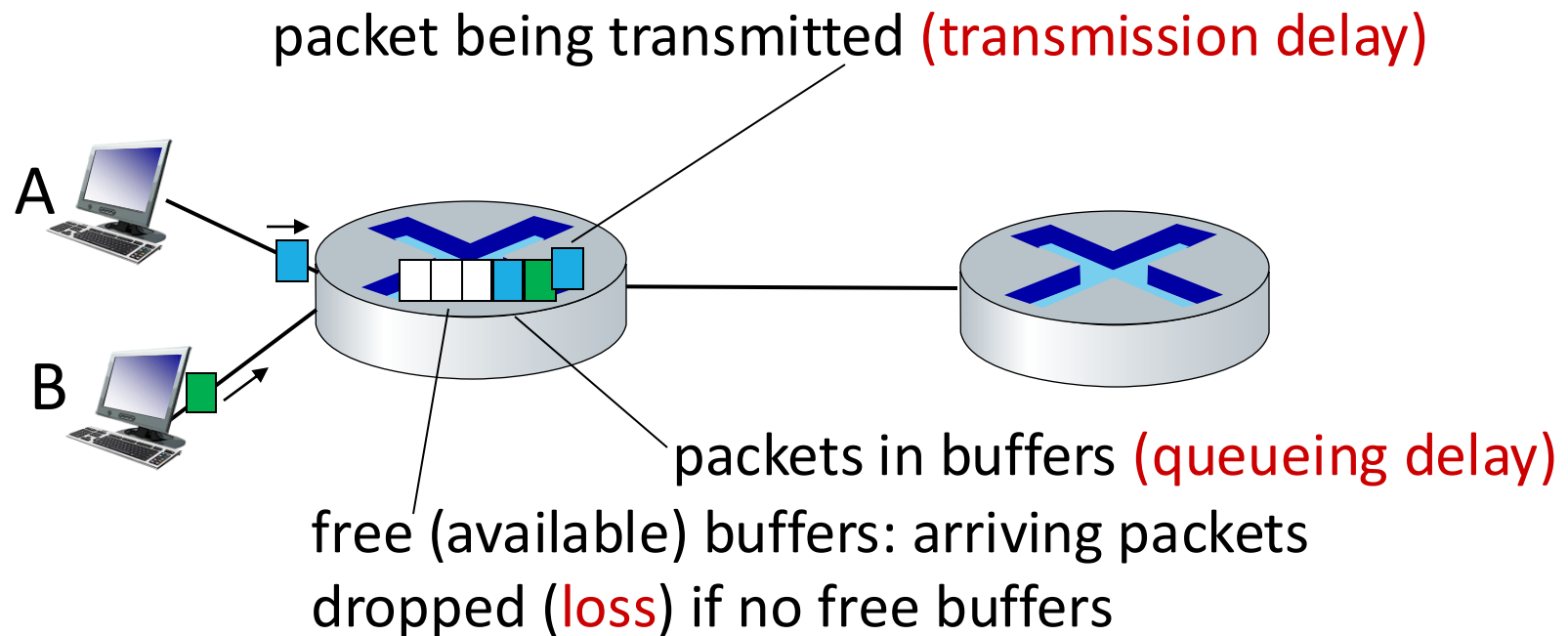
History

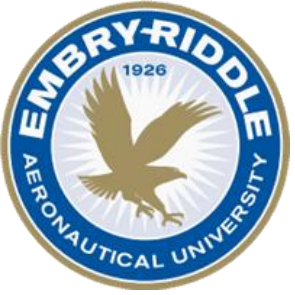


How do packet delay and loss occur?

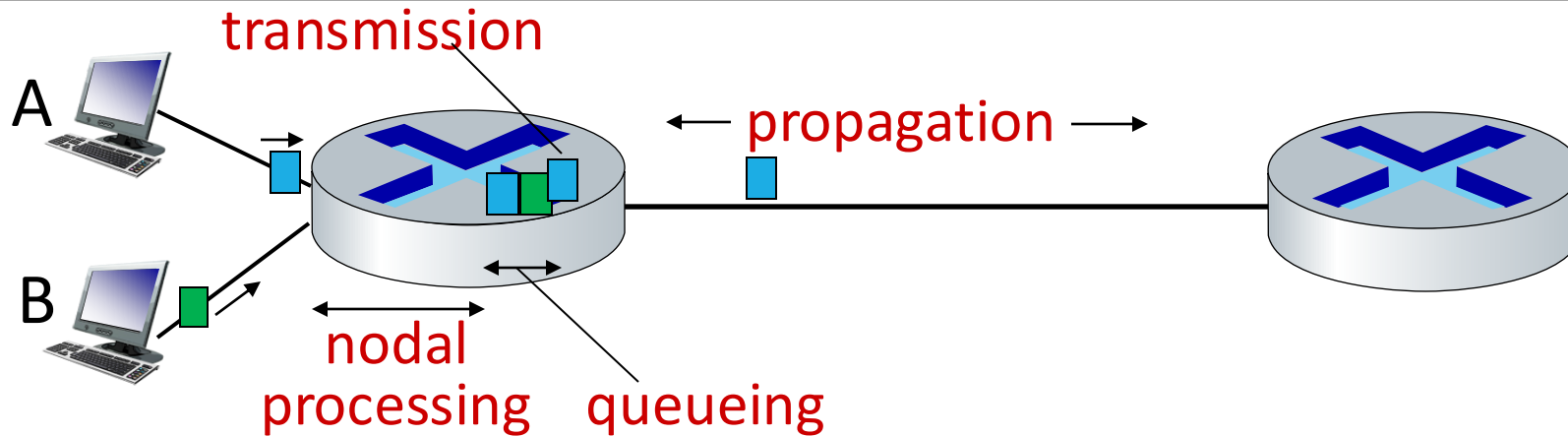
packets *queue* in router buffers, waiting for turn for transmission

- queue length grows when arrival rate to link (temporarily) exceeds output link capacity
- packet *loss* occurs when memory to hold queued packets fills up





Packet delay: four sources



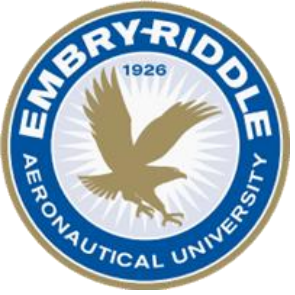
$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{proc} : nodal processing

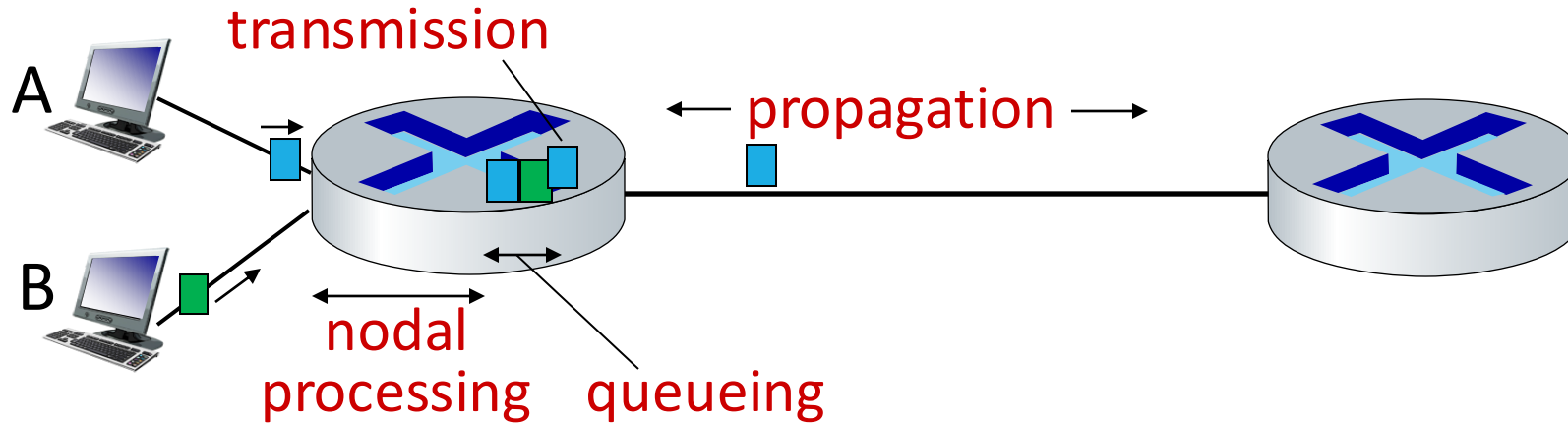
- check bit errors
- determine output link
- typically < microsecs

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router



Packet delay: four sources



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay:

- L : packet length (bits)
- R : link transmission rate (bps)

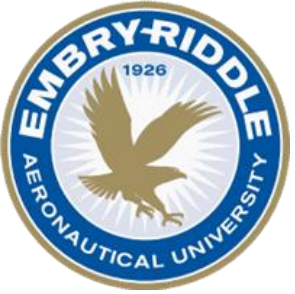
■ $d_{\text{trans}} = L/R$

d_{trans} and d_{prop}
very different

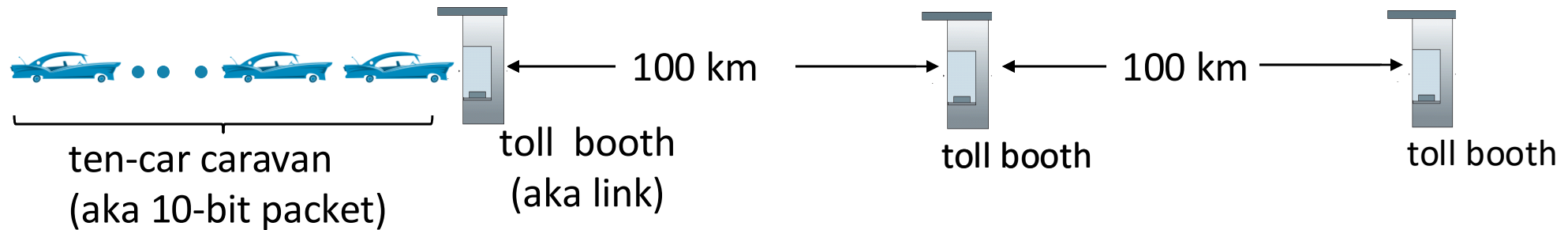
d_{prop} : propagation delay:

- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)

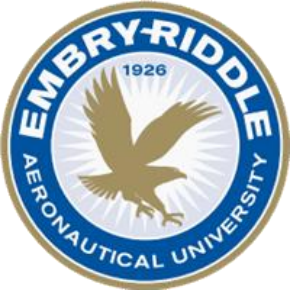
■ $d_{\text{prop}} = d/s$



Caravan analogy



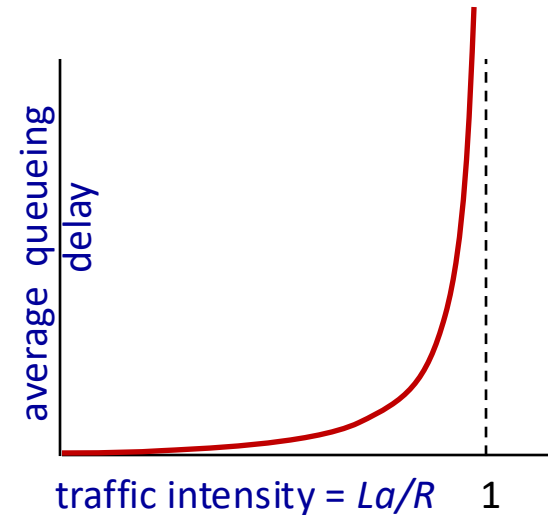
- car $\sim$ bit; caravan $\sim$ packet; toll service $\sim$ link transmission
- toll booth takes 12 sec to service car (bit transmission time)
- “propagate” at 100 km/hr
- **Q: How long until caravan is lined up before 2nd toll booth?**
- time to “push” entire caravan through toll booth onto highway = $12 * 10 = 120$ sec
- time for last car to propagate from 1st to 2nd toll booth: $100\text{km} / (100\text{km/hr}) = 1$ hr
- **A: 62 minutes**



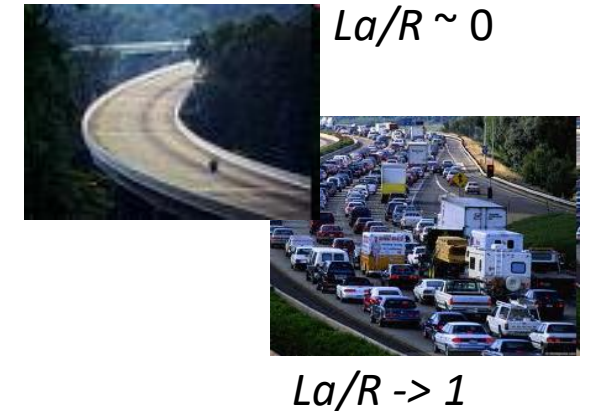
Packet queueing delay

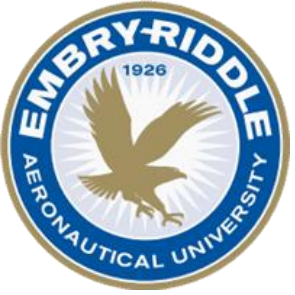
- a : average packet arrival rate
- L : packet length (bits)
- R : link bandwidth (bit transmission rate)

$$\frac{L \cdot a}{R} : \frac{\text{arrival rate of bits}}{\text{service rate of bits}} \quad \text{“traffic intensity”}$$



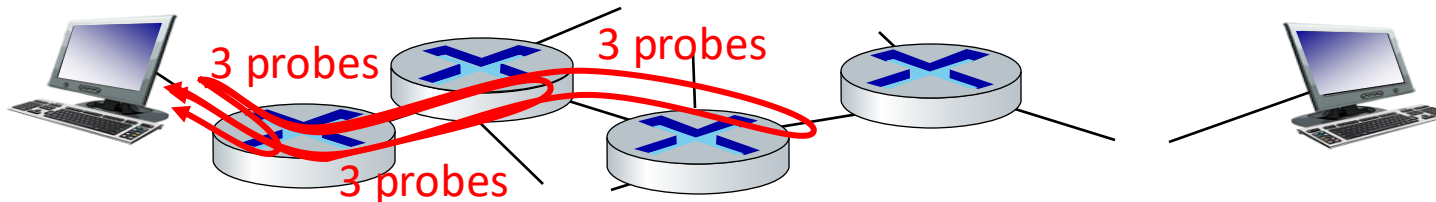
- $La/R \sim 0$: avg. queueing delay small
- $La/R \rightarrow 1$: avg. queueing delay large
- $La/R > 1$: more “work” arriving is more than can be serviced - average delay infinite!

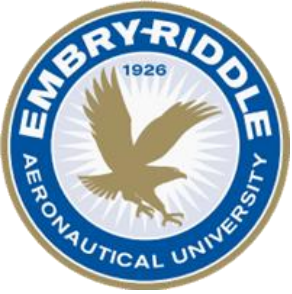




“Real” Internet delays and routes

- what do “real” Internet delay & loss look like?
- **tracert** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
 - router i will return packets to sender
 - sender measures time interval between transmission and reply





Real Internet delays and routes

tracert (on Mac terminal)
tracert (on Windows command line)
tracert (on a Linux command line)

tracert: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
```

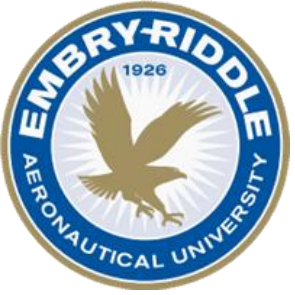
trans-oceanic link

looks like delays decrease! Why?

* means no response (probe lost, router not replying)

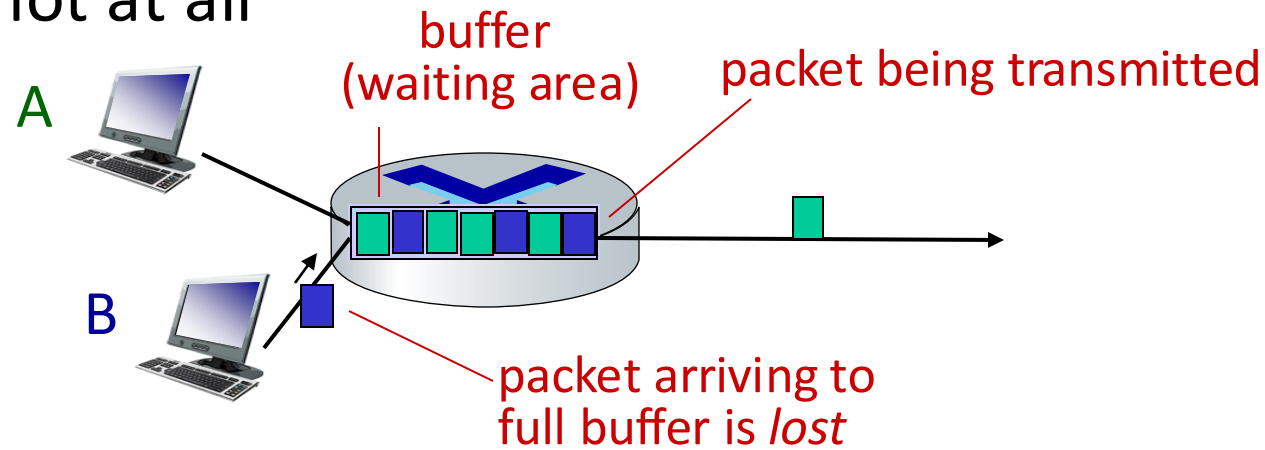
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms

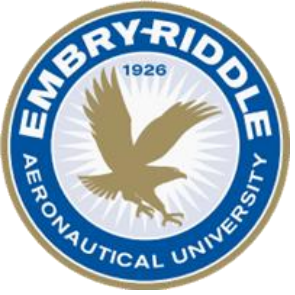
* Do some traceroutes from exotic countries at www.traceroute.org



Packet loss

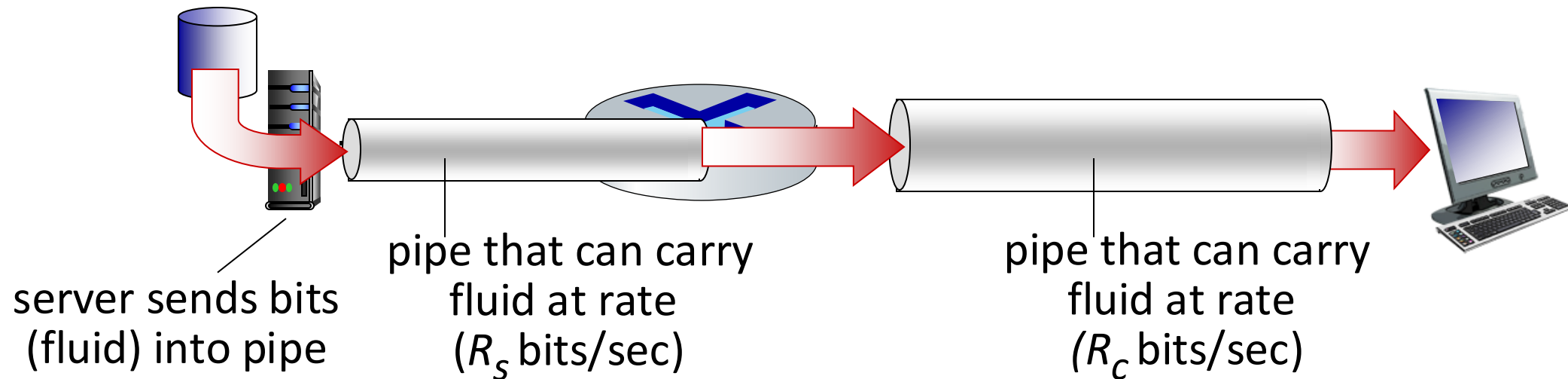
- queue (buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all

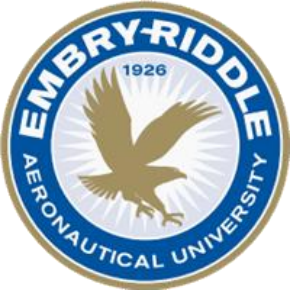




Throughput

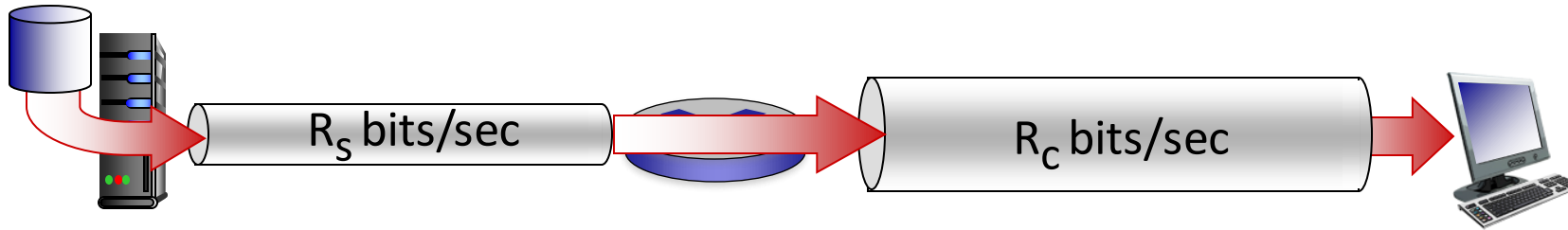
- *throughput*: rate (bits/time unit) at which bits are being sent from sender to receiver
 - *instantaneous*: rate at given point in time
 - *average*: rate over longer period of time



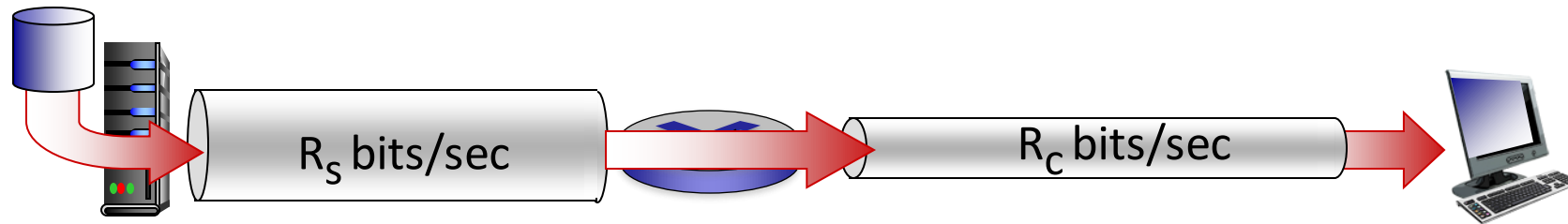


Throughput

$R_s < R_c$ What is average end-end throughput?

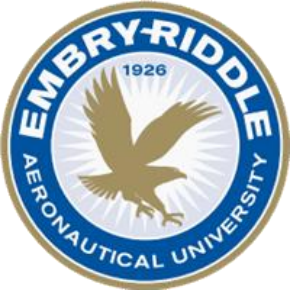


$R_s > R_c$ What is average end-end throughput?

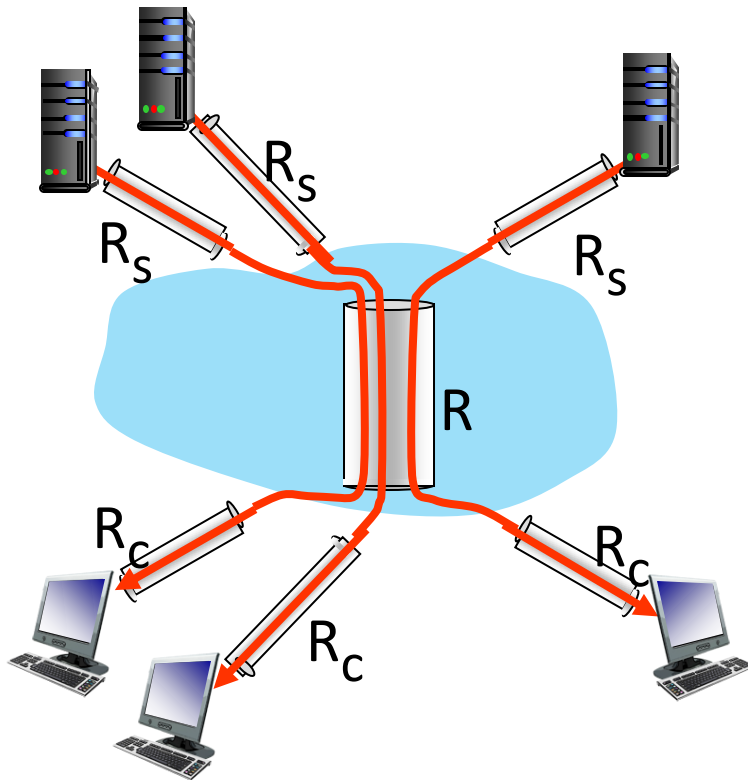


bottleneck link

link on end-end path that constrains end-end throughput

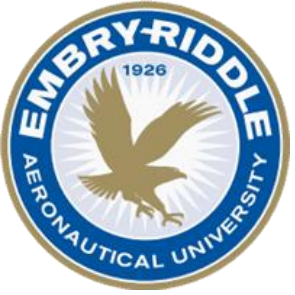


Throughput: network scenario



- per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

10 connections (fairly) share backbone
bottleneck link R bits/sec



Roadmap

What *is* the Internet?

What *is* a protocol?

Network edge: hosts, access network,
physical media

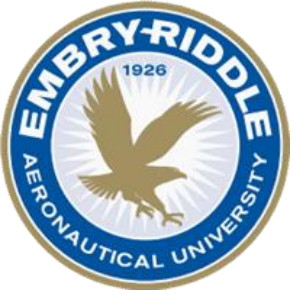
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Performance: loss, delay, throughput

Security

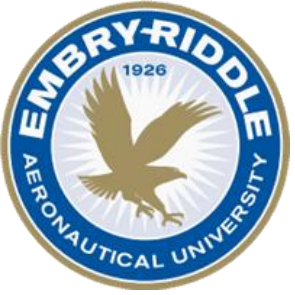
Protocol layers, service models

History



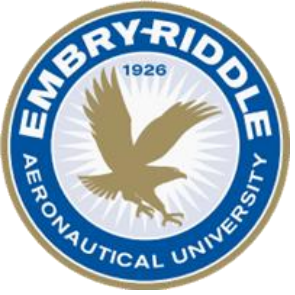
Network security

- Internet not originally designed with (much) security in mind
 - *original vision*: “a group of mutually trusting users attached to a transparent network”
 - Internet protocol designers playing “catch-up”
 - security considerations in all layers!



Network security

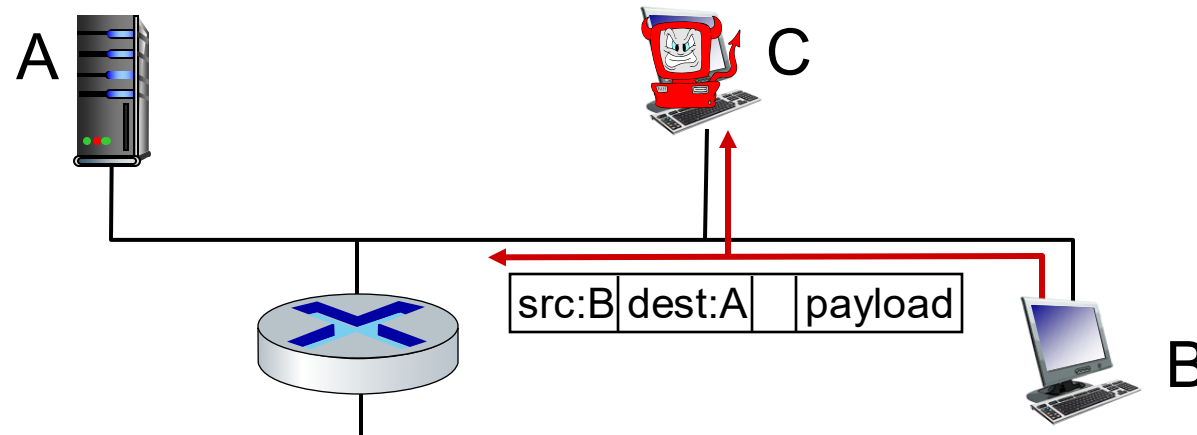
- Internet not originally designed with (much) security in mind
 - *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
 - Internet protocol designers playing “catch-up”
 - security considerations in all layers!
- We now need to think about:
 - how bad guys can attack computer networks
 - how we can defend networks against attacks
 - how to design architectures that are immune to attacks



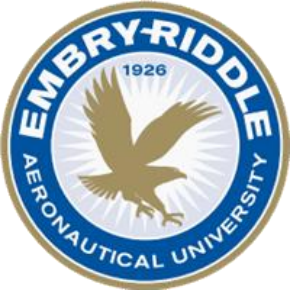
Bad guys: packet interception

packet “sniffing”:

- broadcast media (shared Ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by

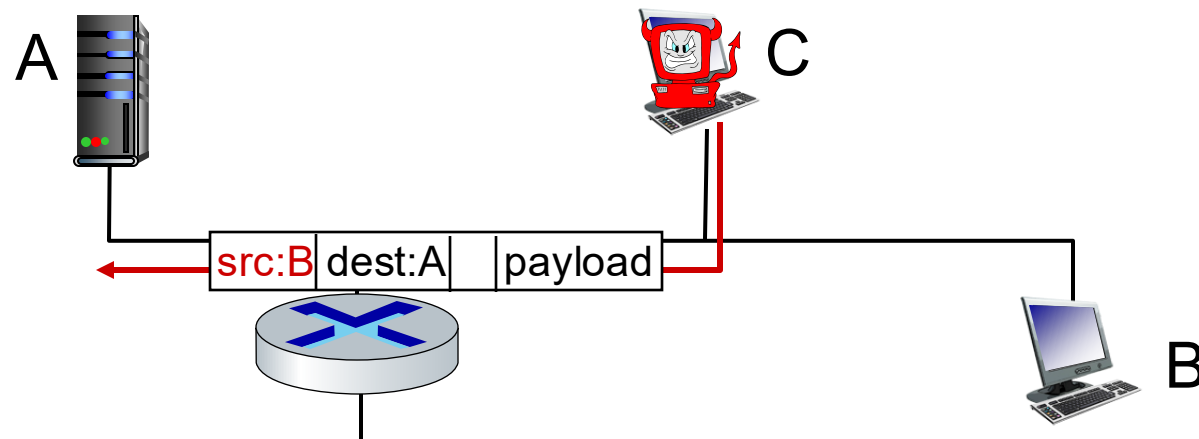


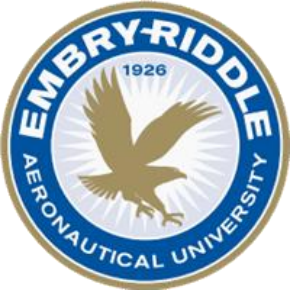
Wireshark software used for our labs is a (free) packet-sniffer



Bad guys: fake identity

IP spoofing: injection of packet with false source address

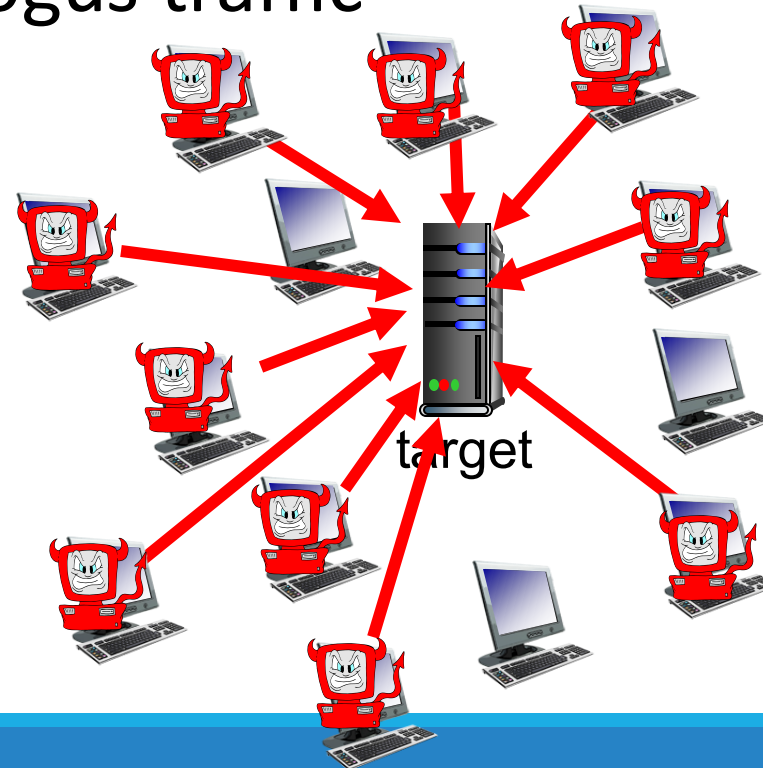


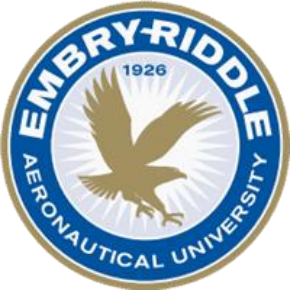


Bad guys: denial of service

Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

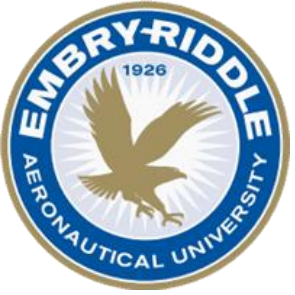
1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts





Lines of defense:

- **authentication:** proving you are who you say you are
 - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- **confidentiality:** via encryption
- **integrity checks:** digital signatures prevent/detect tampering
- **access restrictions:** password-protected VPNs
- **firewalls:** specialized “middleboxes” in access and core networks:
 - off-by-default: filter incoming packets to restrict senders, receivers, applications
 - detecting/reacting to DOS attacks



Roadmap

What *is* the Internet?

What *is* a protocol?

Network edge: hosts, access network,
physical media

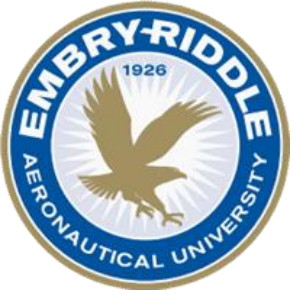
Network core: packet/circuit switching,
internet structure

Performance: loss, delay, throughput

Security

Protocol layers, service models

History



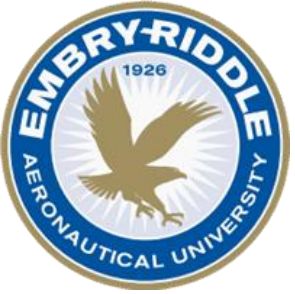
Protocol “layers” and reference models

Networks are complex,
with many “pieces”:

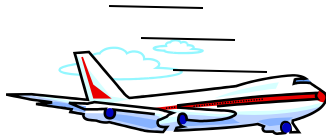
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any
hope of *organizing*
structure of network?

- and/or our *discussion*
of networks?



Example: organization of air travel



end-to-end transfer of person plus baggage

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

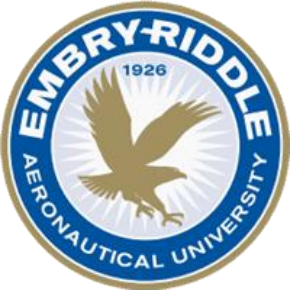
runway landing

airplane routing

airplane routing

How would you *define/discuss* the *system* of airline travel?

- a series of steps, involving many services

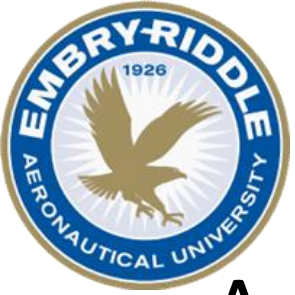


Example: organization of air travel



layers: each layer implements a service

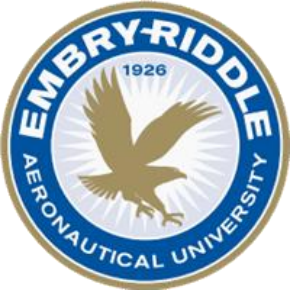
- via its own internal-layer actions
- relying on services provided by layer below



Why layering?

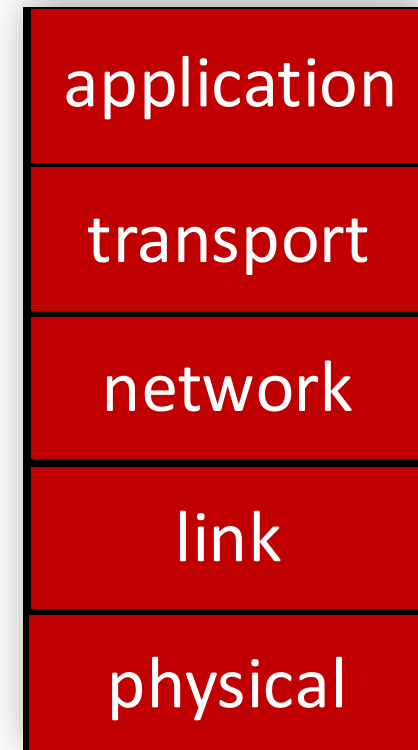
Approach to designing/discussing complex systems:

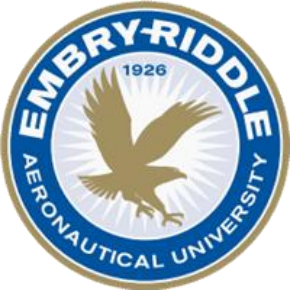
- explicit structure allows identification, relationship of system's pieces
 - layered *reference model* for discussion
- modularization eases maintenance, updating of system
 - change in layer's service *implementation*: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system



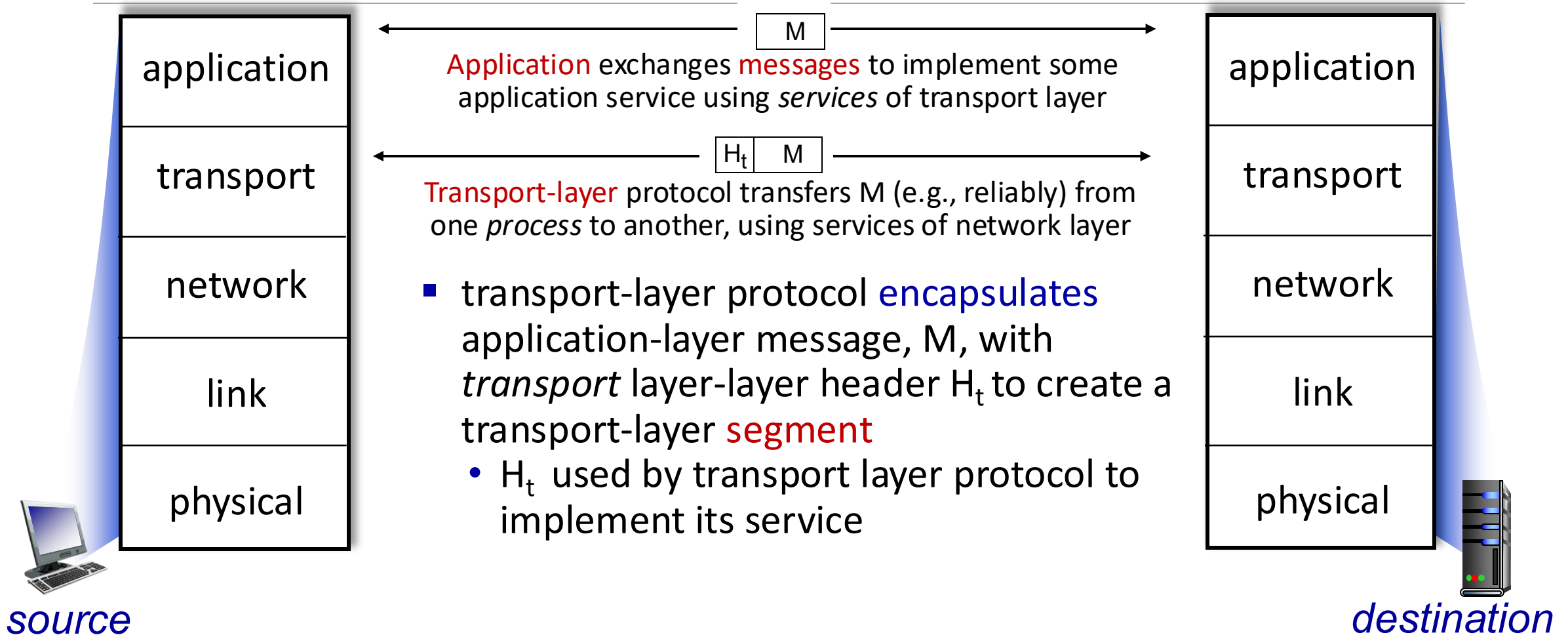
Layered Internet protocol stack

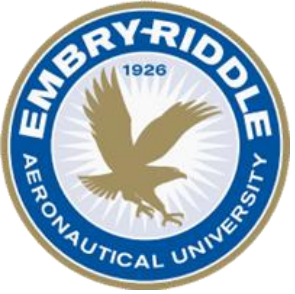
- *application*: supporting network applications
 - HTTP, IMAP, SMTP, DNS
- *transport*: process-process data transfer
 - TCP, UDP
- *network*: routing of datagrams from source to destination
 - IP, routing protocols
- *link*: data transfer between neighboring network elements
 - Ethernet, 802.11 (WiFi), PPP
- *physical*: bits “on the wire”



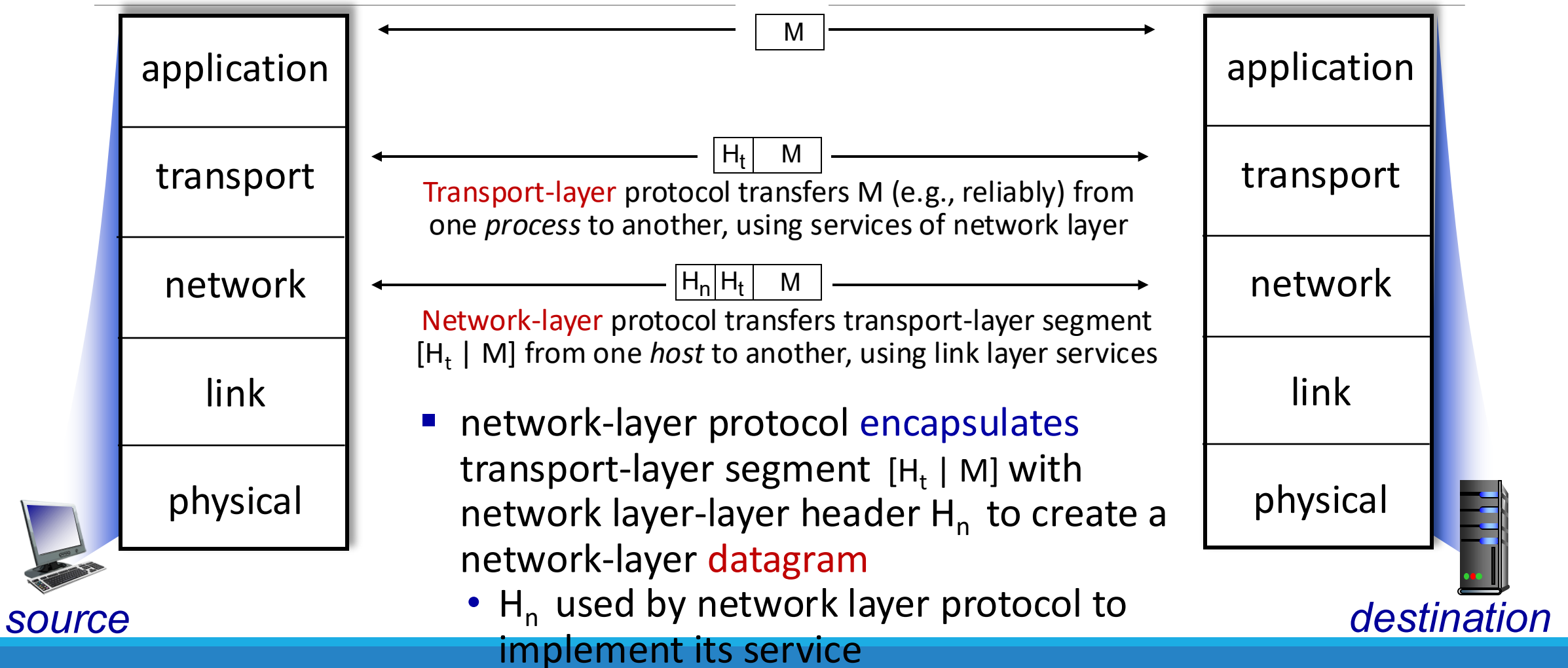


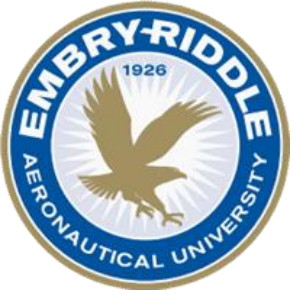
Services, Layering and Encapsulation



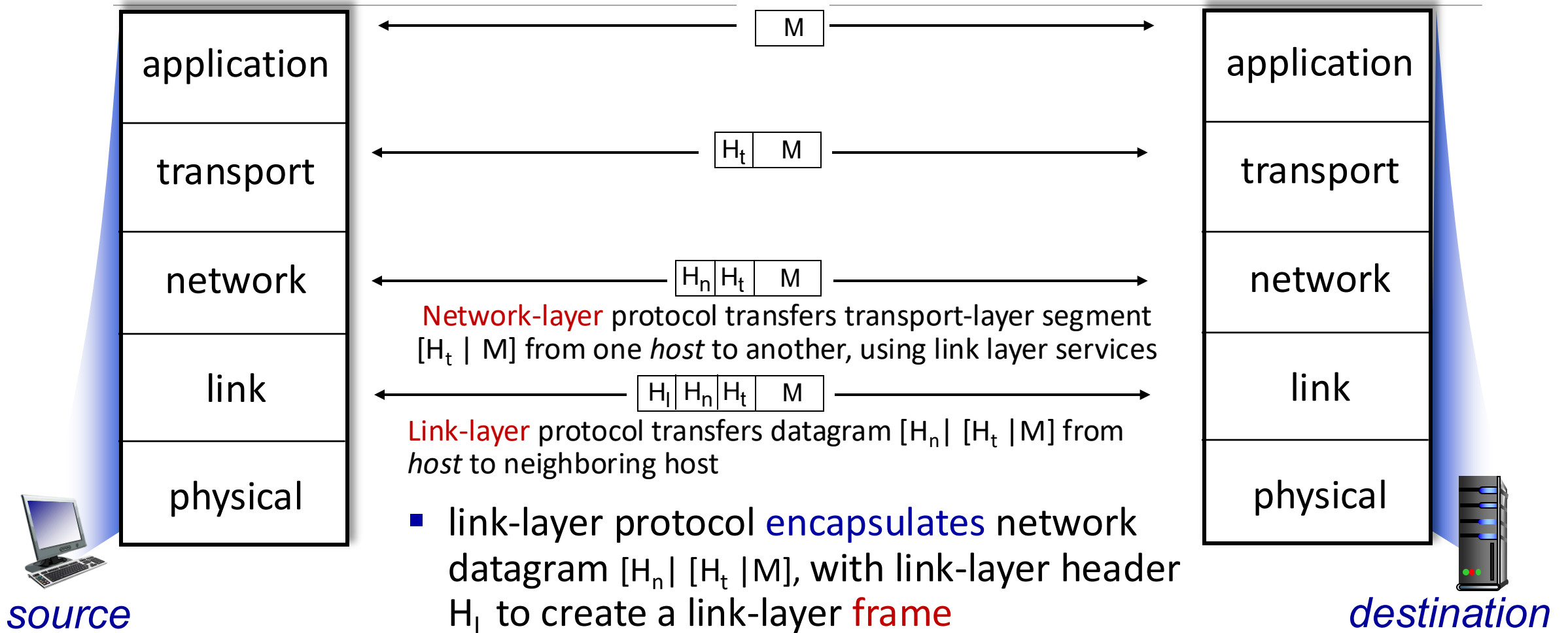


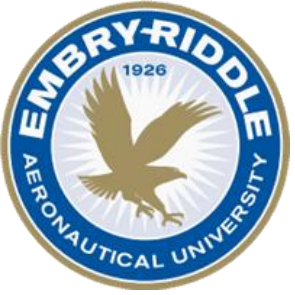
Services, Layering and Encapsulation



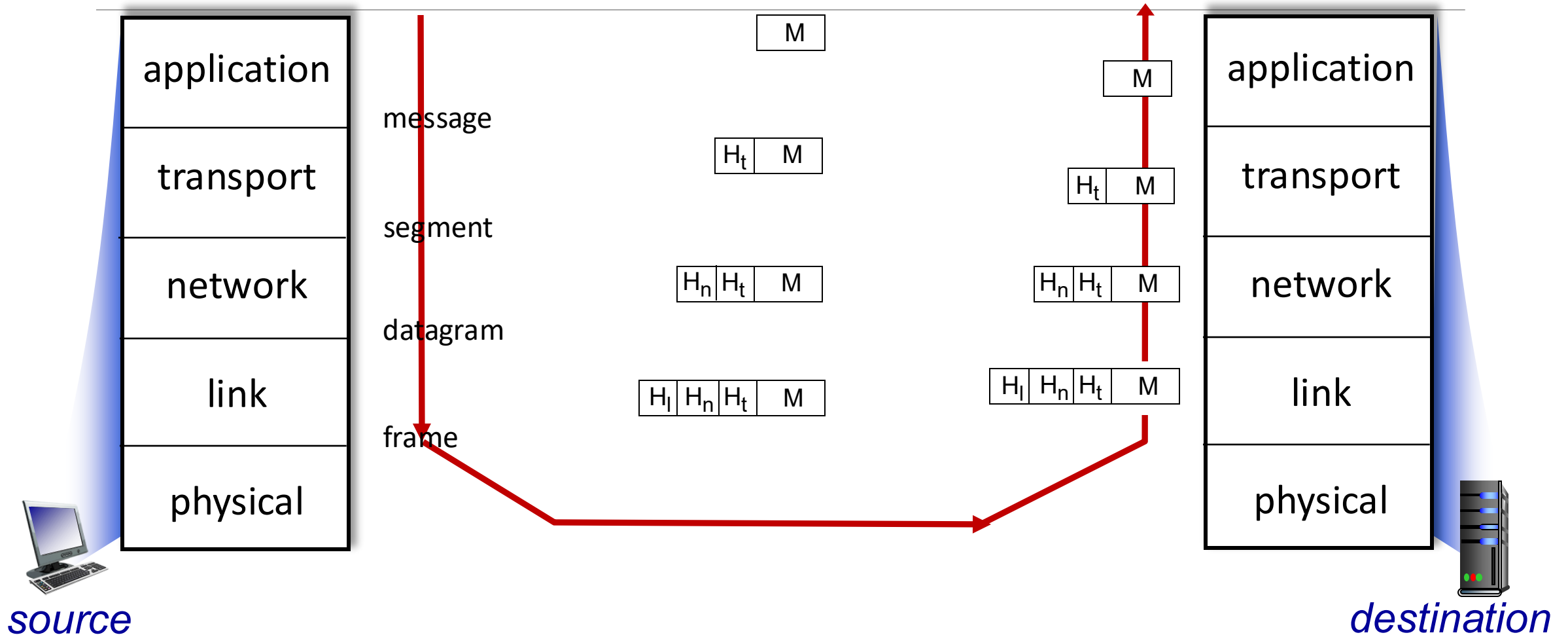


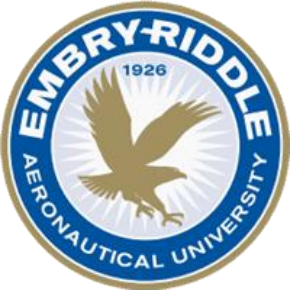
Services, Layering and Encapsulation



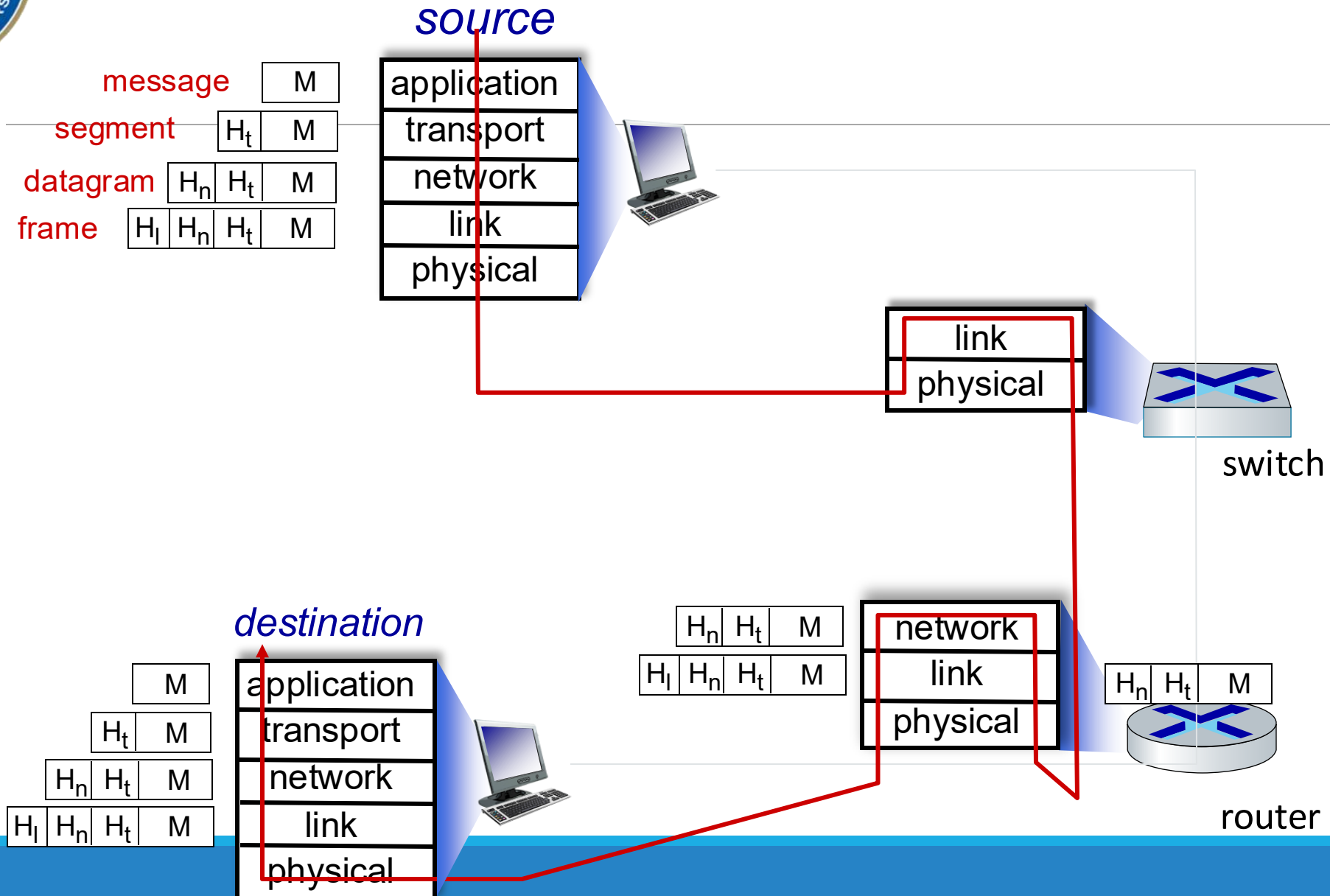


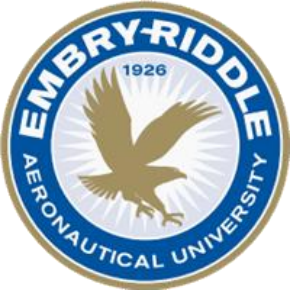
Services, Layering and Encapsulation





Encapsulation: an end-end view

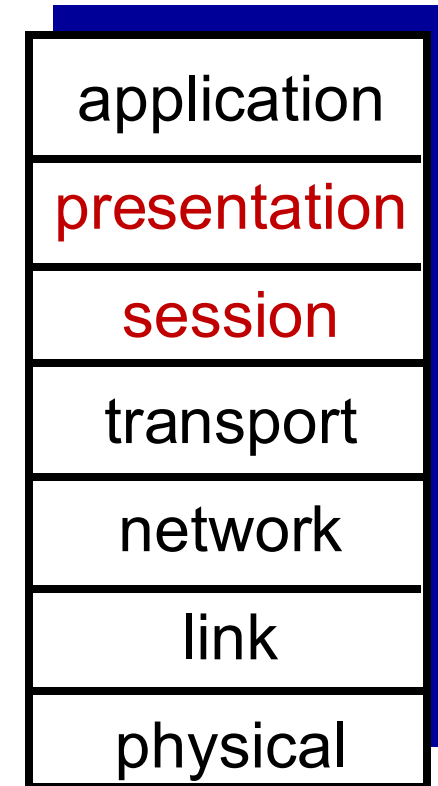




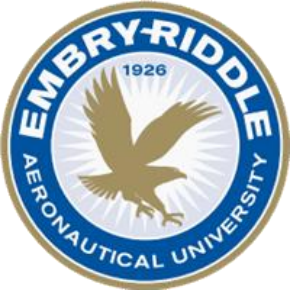
ISO/OSI reference model

Two layers not found in Internet protocol stack!

- *presentation*: allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- *session*: synchronization, checkpointing, recovery of data exchange
- Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in application
 - needed?



The seven layer OSI/ISO reference model



Roadmap

What *is* the Internet?

What *is* a protocol?

Network edge: hosts, access network,
physical media

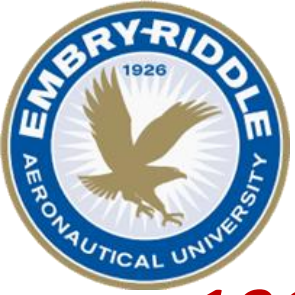
Network core: packet/circuit switching,
internet structure

Performance: loss, delay, throughput

Security

Protocol layers, service models

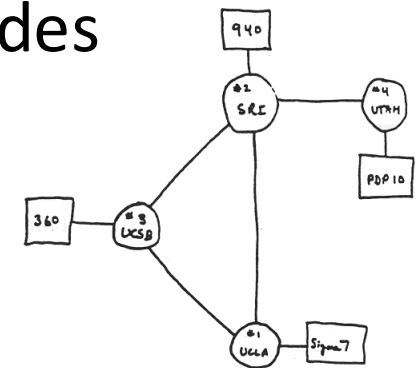
History



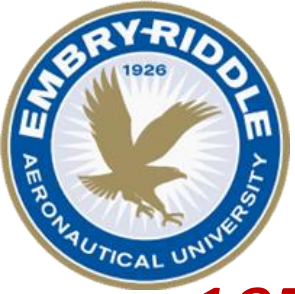
Internet history

1961-1972: Early packet-switching principles

- **1961:** Kleinrock - queueing theory shows effectiveness of packet-switching
- **1964:** Baran - packet-switching in military nets
- **1967:** ARPAnet conceived by Advanced Research Projects Agency
- **1969:** first ARPAnet node operational
- **1972:**
 - ARPAnet public demo
 - NCP (Network Control Protocol) first host-host protocol
 - first e-mail program
 - ARPAnet has 15 nodes



THE ARPA NETWORK



Internet history

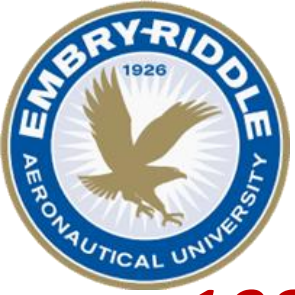
1972-1980: Internetworking, new and proprietary networks

- **1970:** ALOHAnet satellite network in Hawaii
- **1974:** Cerf and Kahn - architecture for interconnecting networks
- **1976:** Ethernet at Xerox PARC
- **late 70's:** proprietary architectures: DECnet, SNA, XNA
- **1979:** ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best-effort service model
- stateless routing
- decentralized control

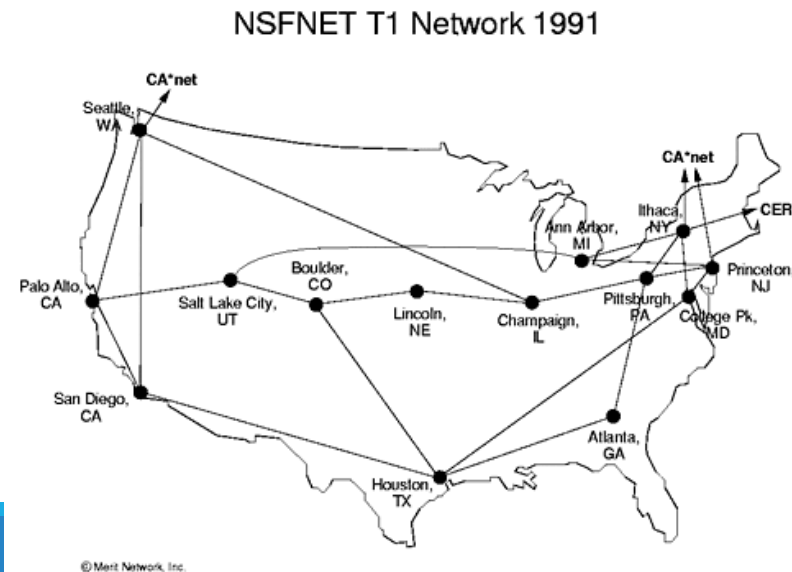
define today's Internet architecture

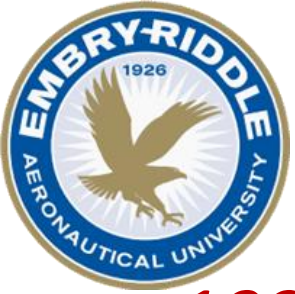


Internet history

1980-1990: new protocols, a proliferation of networks

- 1983: deployment of TCP/IP
- 1982: smtp e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: ftp protocol defined
- 1988: TCP congestion control
- new national networks: CSnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks

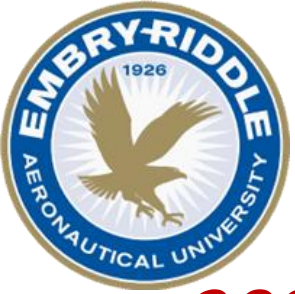




Internet history

1990, 2000s: commercialization, the Web, new applications

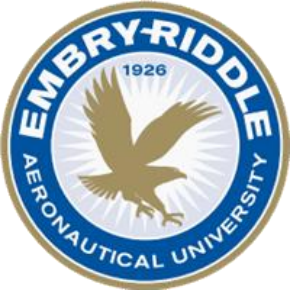
- early 1990s: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- early 1990s: Web
 - hypertext [Bush 1945, Nelson 1960's]
 - HTML, HTTP: Berners-Lee
 - 1994: Mosaic, later Netscape
 - late 1990s: commercialization of the Web
- late 1990s – 2000s:
 - more killer apps: instant messaging, P2P file sharing
 - network security to forefront
 - est. 50 million host, 100 million+ users
 - backbone links running at Gbps



Internet history

2005-present: scale, SDN, mobility, cloud

- aggressive deployment of broadband home access (10-100's Mbps)
- 2008: software-defined networking (SDN)
- increasing ubiquity of high-speed wireless access: 4G/5G, WiFi
- service providers (Google, FB, Microsoft) create their own networks
 - bypass commercial Internet to connect “close” to end user, providing “instantaneous” access to social media, search, video content, ...
- enterprises run their services in “cloud” (e.g., Amazon Web Services, Microsoft Azure)
- rise of smartphones: more mobile than fixed devices on Internet (2017)
- ~18B devices attached to Internet (2017)



Summary

We've covered a "ton" of material!

- Internet overview
- what's a protocol?
- network edge, access network, core
 - packet-switching versus circuit-switching
 - Internet structure
- performance: loss, delay, throughput
- layering, service models
- security
- history

You now have:

- context, overview, vocabulary, "feel" of networking
- more depth, detail, *and fun* to follow!